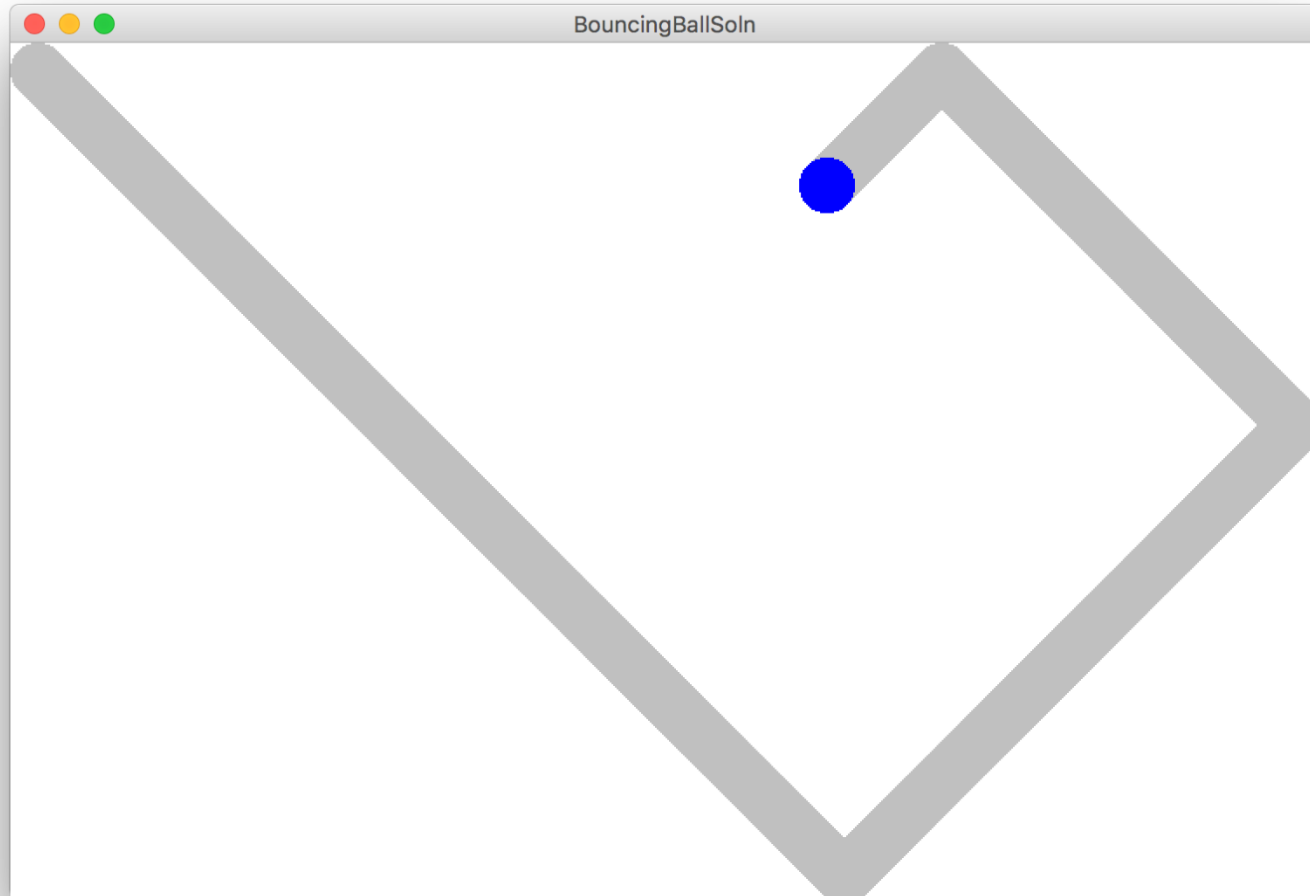


Learning Goals

1. Write animated programs



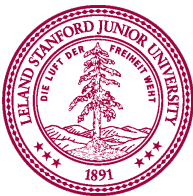
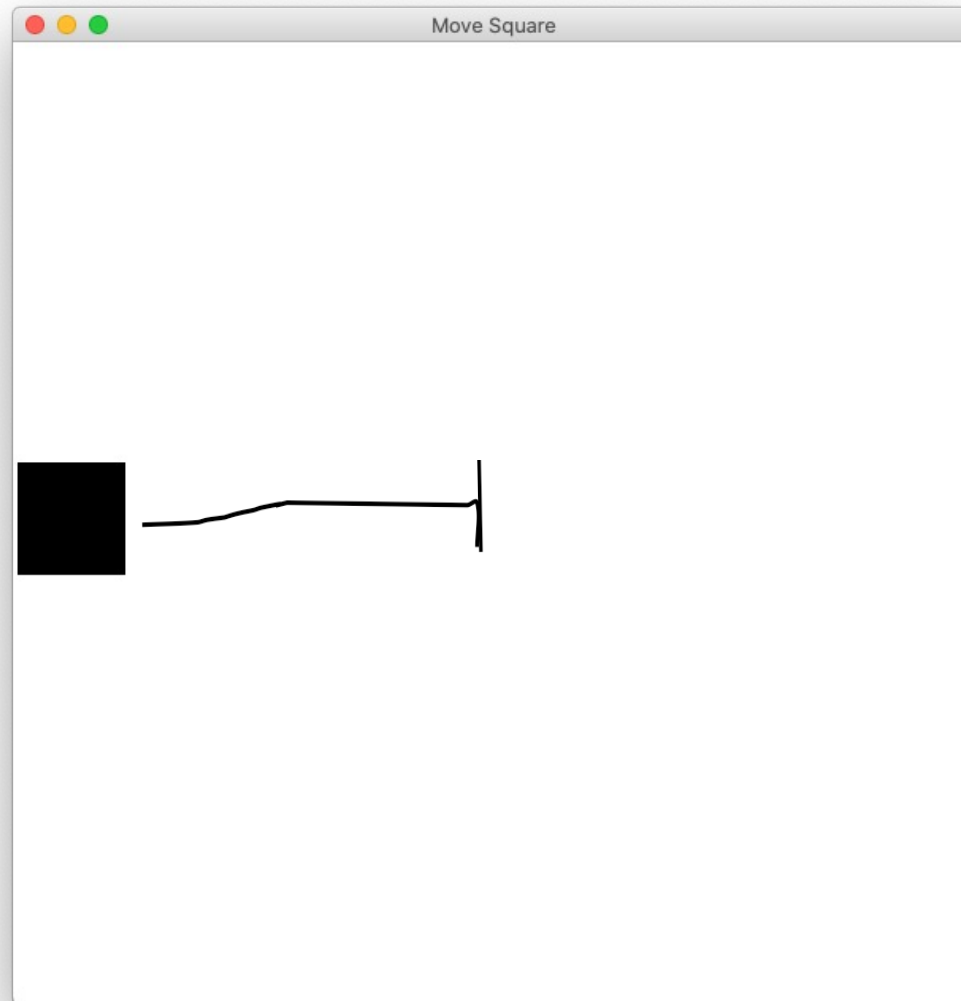
You will be able to write Bouncing Ball



Great foundation



Move to Center

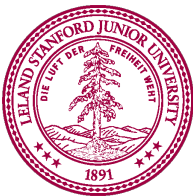


In our last episode...

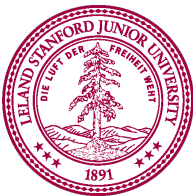
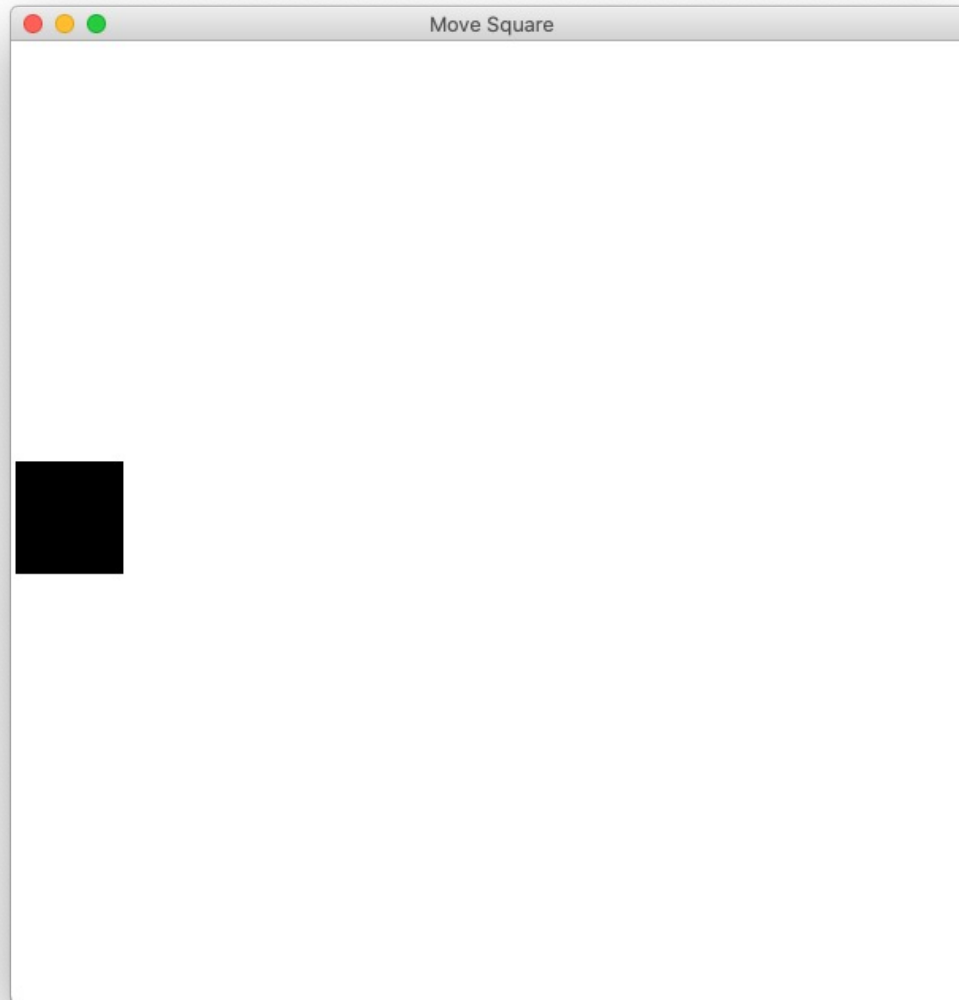
Graphics

```
from graphics import Canvas
```

```
# we usually write this for you, and include  
# it in your projects!  
canvas = Canvas(width, height)
```



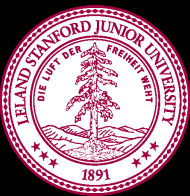
Add square



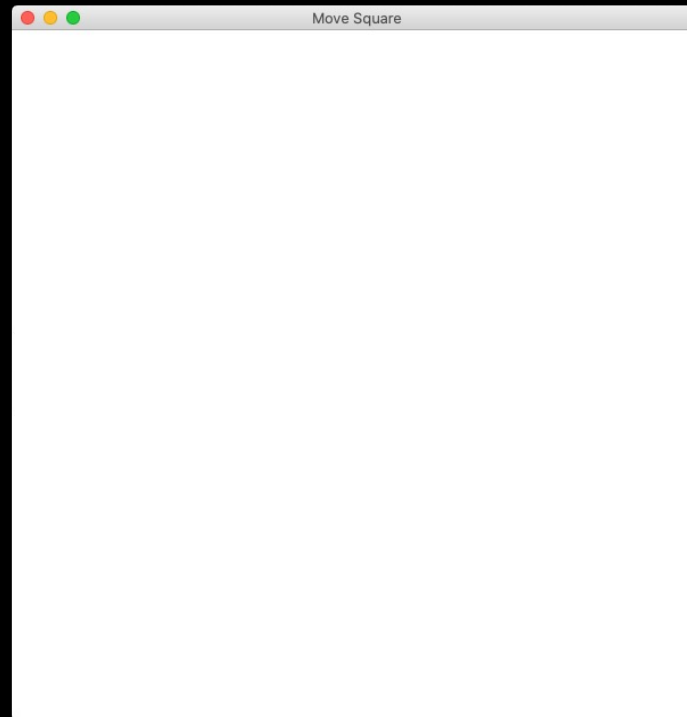
```
def main():  
    canvas = Canvas(CANVAS_WIDTH, CANVAS_HEIGHT)  
    start_x = 0  
    start_y = CANVAS_HEIGHT / 2 - SQUARE_SIZE / 2  
    square = canvas.create_rectangle(start_x, start_y,  
                                     SQUARE_SIZE, start_y + SQUARE_SIZE)
```



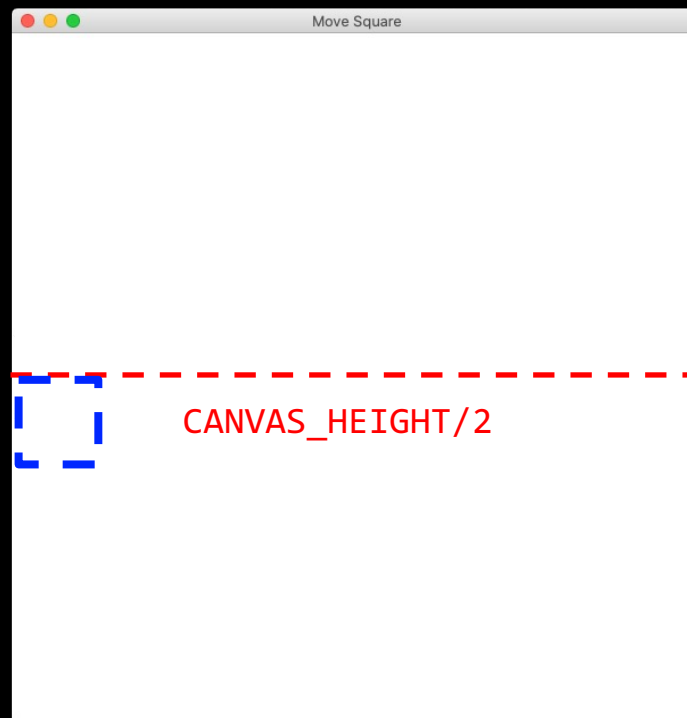

```
def main():  
    canvas = Canvas(CANVAS_WIDTH, CANVAS_HEIGHT)  
    start_x = 0  
    start_y = CANVAS_HEIGHT / 2 - SQUARE_SIZE / 2  
    square = canvas.create_rectangle(start_x, start_y,  
                                     SQUARE_SIZE, start_y + SQUARE_SIZE)
```



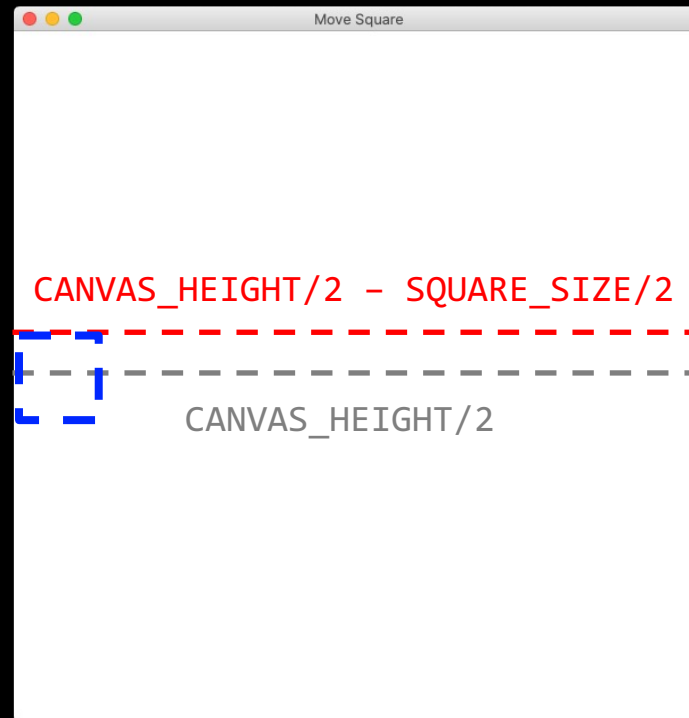
```
def main():  
    canvas = Canvas(CANVAS_WIDTH, CANVAS_HEIGHT)  
    start_x = 0  
    start_y = CANVAS_HEIGHT / 2 - SQUARE_SIZE / 2  
    square = canvas.create_rectangle(start_x, start_y,  
                                     SQUARE_SIZE, start_y + SQUARE_SIZE)
```



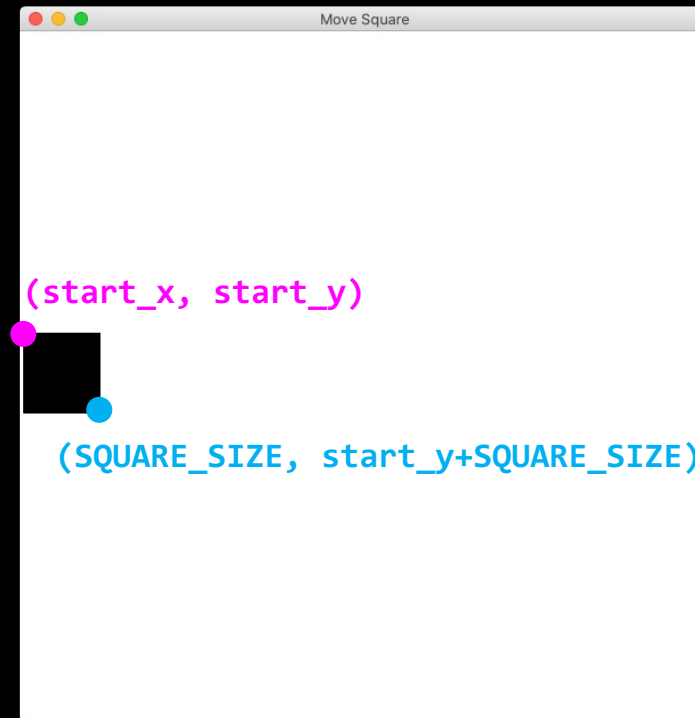
```
def main():  
    canvas = Canvas(CANVAS_WIDTH, CANVAS_HEIGHT)  
    start_x = 0  
    start_y = CANVAS_HEIGHT / 2 - SQUARE_SIZE / 2  
    square = canvas.create_rectangle(start_x, start_y,  
                                     SQUARE_SIZE, start_y + SQUARE_SIZE)
```



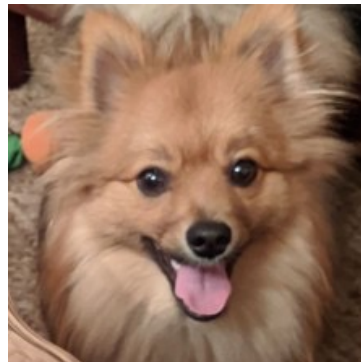
```
def main():  
    canvas = Canvas(CANVAS_WIDTH, CANVAS_HEIGHT)  
    start_x = 0  
    start_y = CANVAS_HEIGHT / 2 - SQUARE_SIZE / 2  
    square = canvas.create_rectangle(start_x, start_y,  
                                     SQUARE_SIZE, start_y + SQUARE_SIZE)
```



```
def main():  
    canvas = Canvas(CANVAS_WIDTH, CANVAS_HEIGHT)  
    start_x = 0  
    start_y = CANVAS_HEIGHT / 2 - SQUARE_SIZE / 2  
    square = canvas.create_rectangle(start_x, start_y,  
                                     SQUARE_SIZE, start_y + SQUARE_SIZE)
```



We got the square!

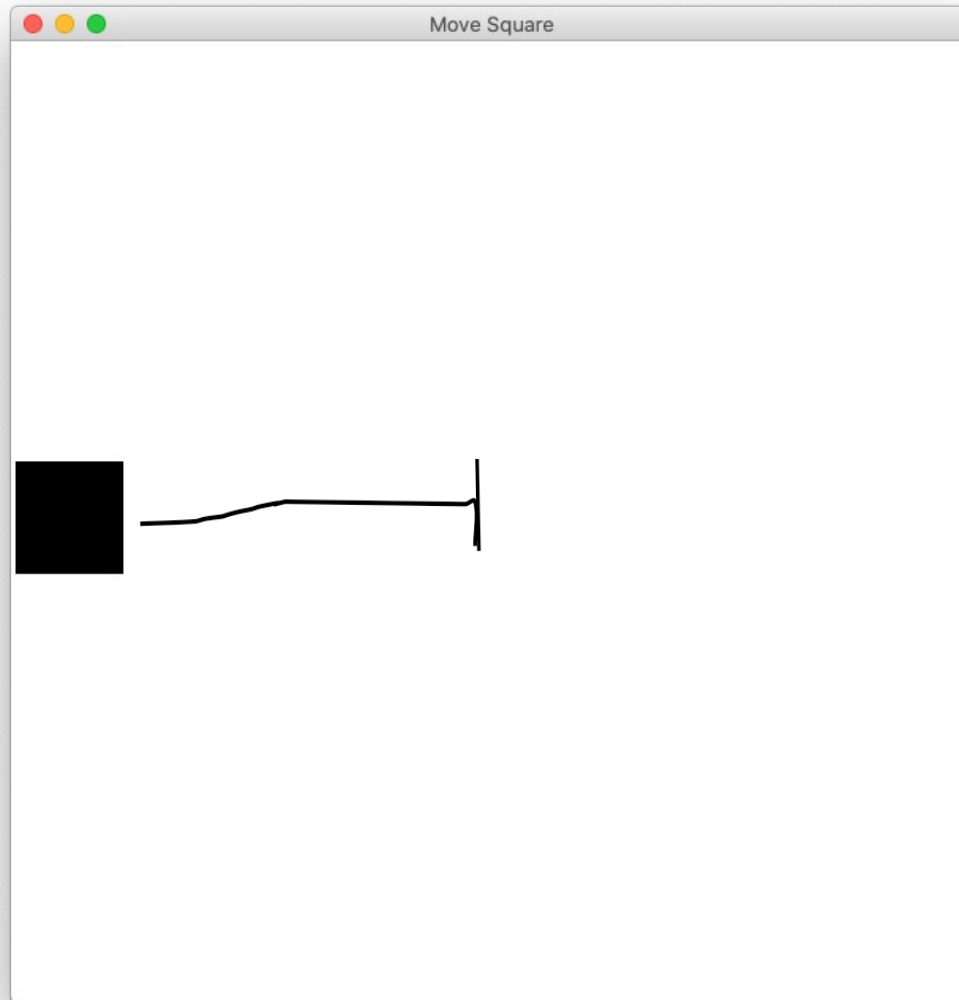


Woot!

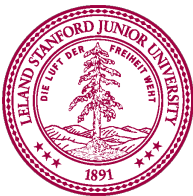
Um... What about the animation?!

How do movies or games
animate?

Move to Center

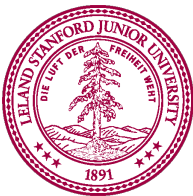


* That's not quite Disney, but it is a start...



Animation Loop

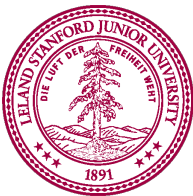
```
def main():  
    # setup  
  
    while True:  
        # update world  
  
        # pause  
        time.sleep(DELAY)
```



Animation Loop

```
def main():  
    # setup  
  
    while True:  
        # update world  
  
        # pause  
        time.sleep(DELAY)
```

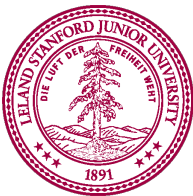
Make all the variables
you need.



Animation Loop

```
def main():  
    # setup  
    while True:  
        # update world  
  
        # pause  
        time.sleep(DELAY)
```

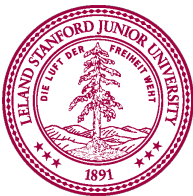
The animation loop is a repetition of heartbeats



Animation Loop

```
def main():  
    # setup  
  
    while True:  
        # update world  
        # pause  
        time.sleep(DELAY)
```

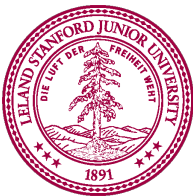
Each heart-beat, update
the world forward one
frame



Animation Loop

```
def main():  
    # setup  
  
    while True:  
        # update world  
        # pause  
        time.sleep(DELAY)
```

If you don't pause,
humans won't be able
to see it



Moving square in action!

Let's see some code!

Move To Center

```
VELOCITY = 2
```

```
def main():
```

```
    # setup
```

```
    canvas = Canvas(CANVAS_WIDTH, CANVAS_HEIGHT)
```

```
    start_x = 0
```

```
    start_y = CANVAS_HEIGHT / 2 - SQUARE_SIZE / 2
```

```
    square = canvas.create_rectangle(start_x, start_y,  
                                     SQUARE_SIZE, start_y + SQUARE_SIZE)
```

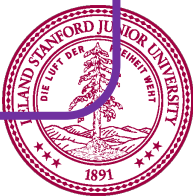
```
    while (start_x + SQUARE_SIZE / 2) < CANVAS_WIDTH / 2:
```

```
        start_x += VELOCITY
```

```
        canvas.moveto(square, start_x, start_y)
```

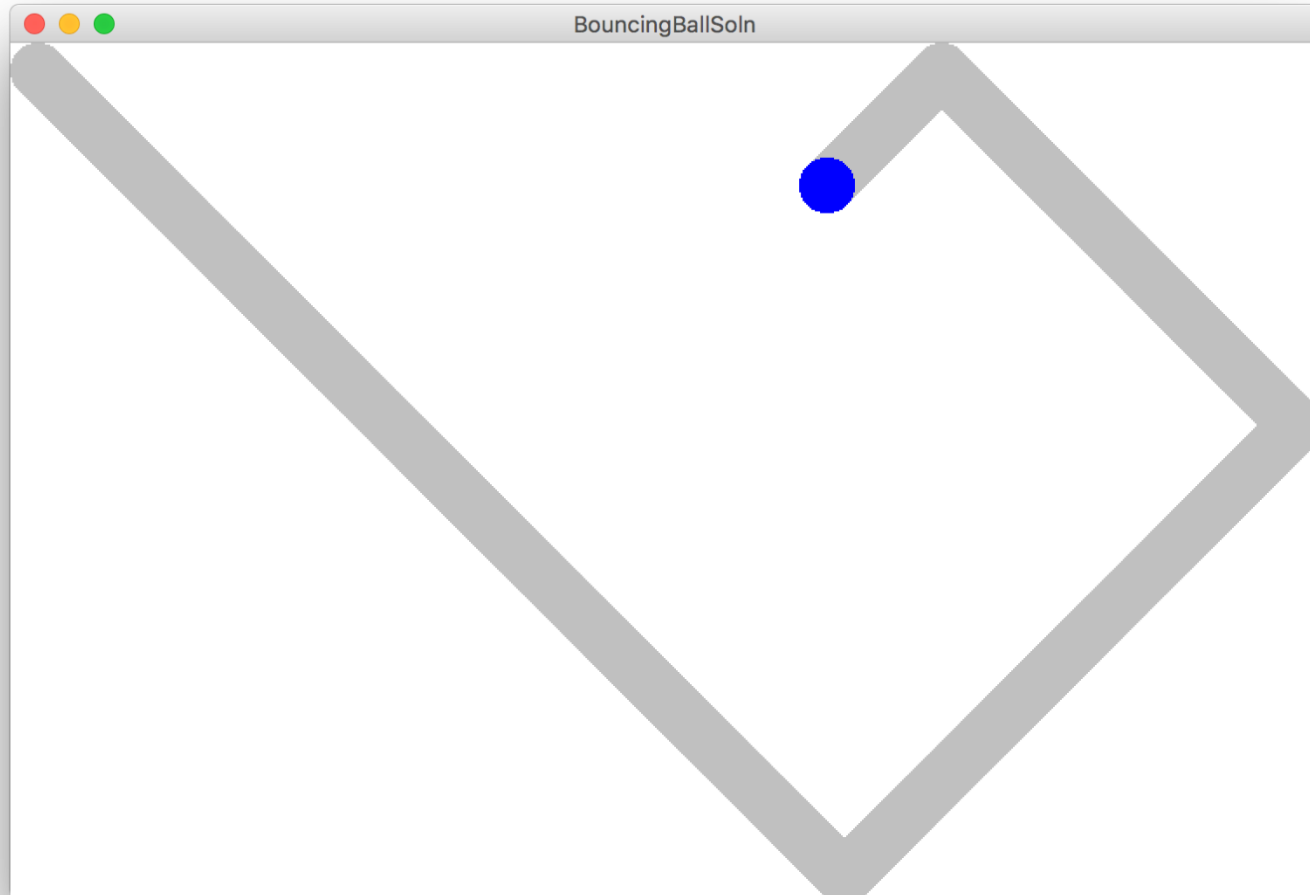
```
    # pause
```

```
    time.sleep(DELAY)
```

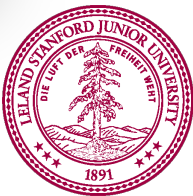
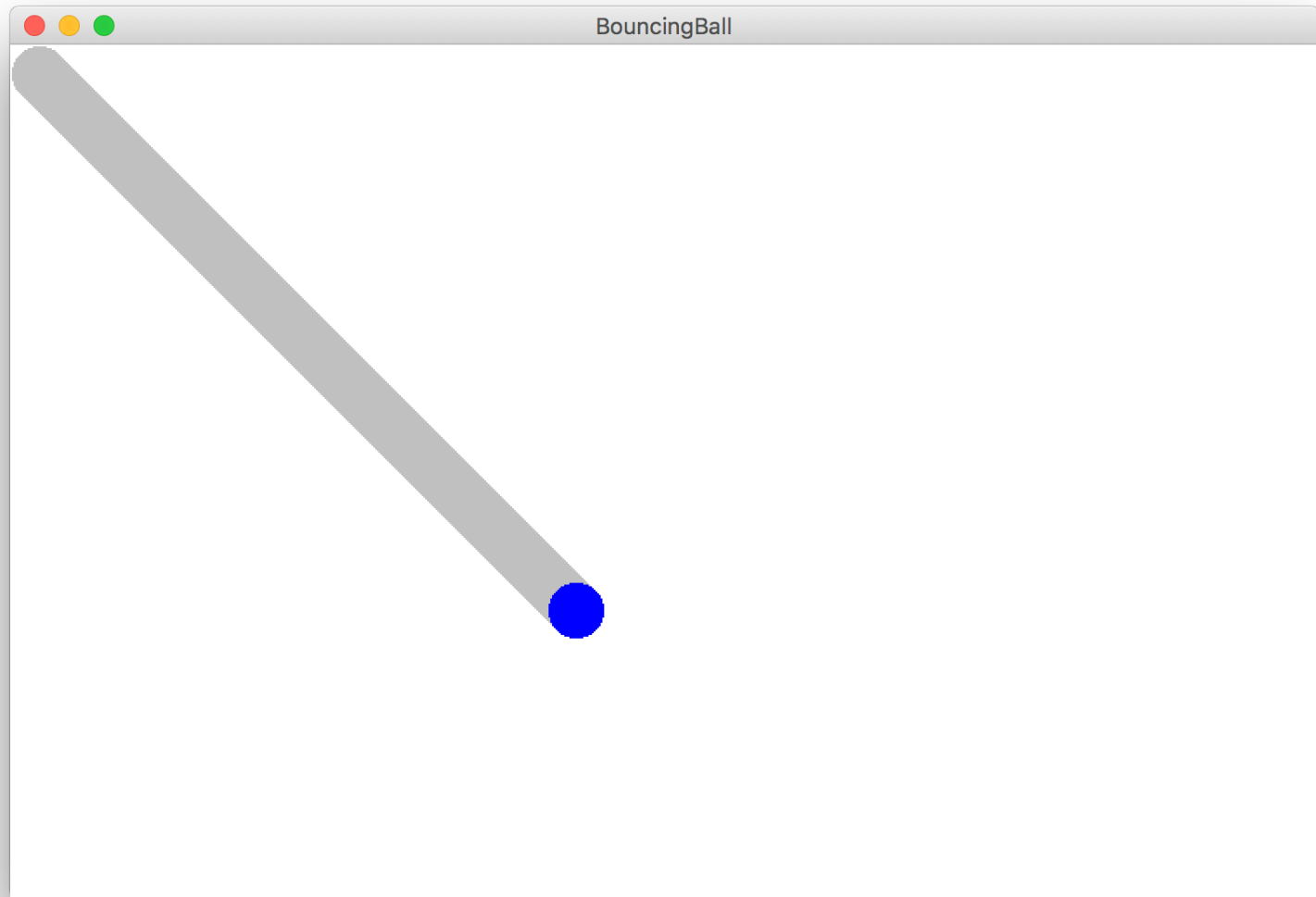


We are ready...

Bouncing Ball

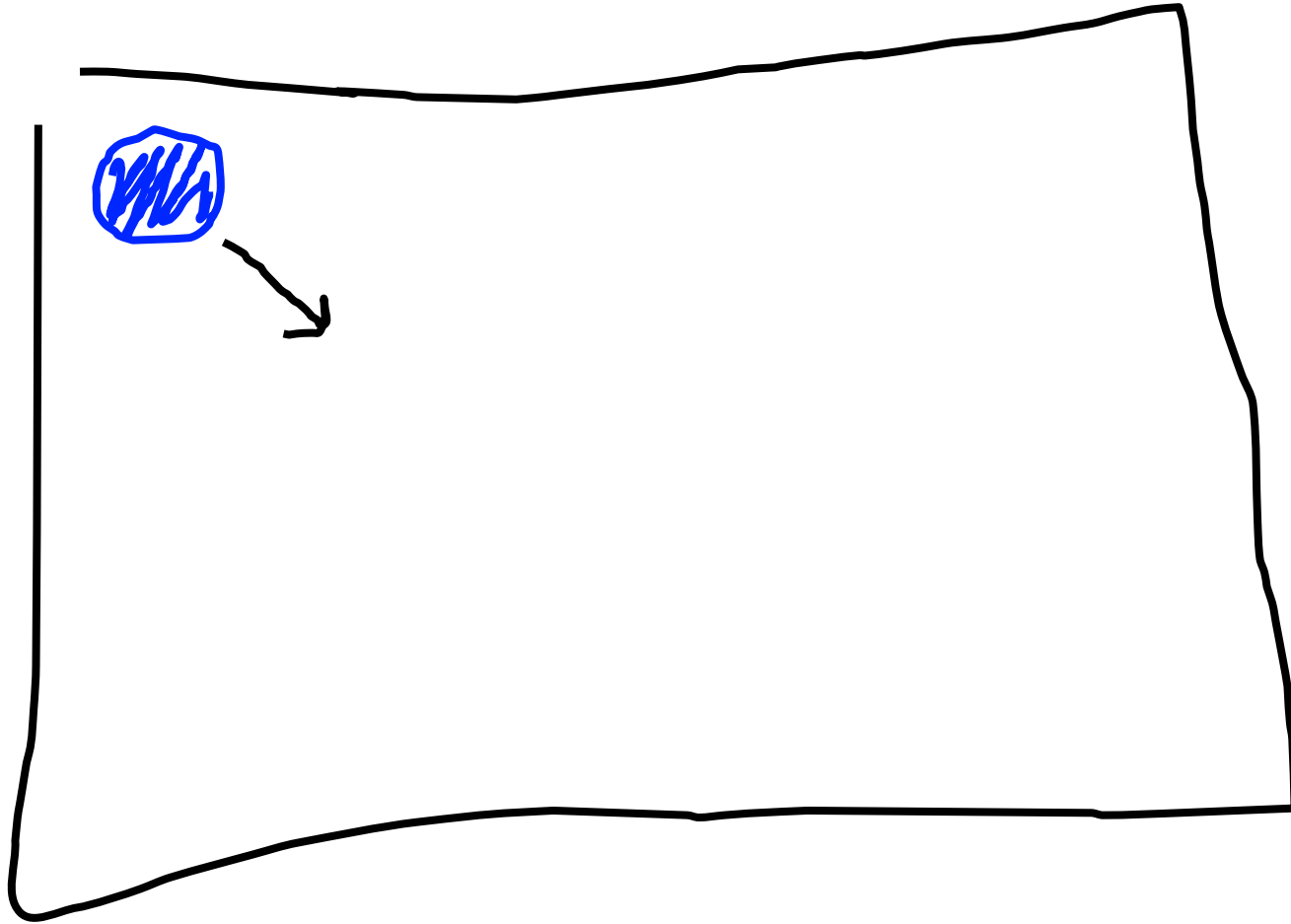


Milestone #1

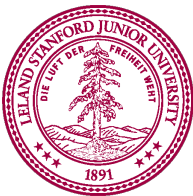


Bouncing Ball

First heartbeat

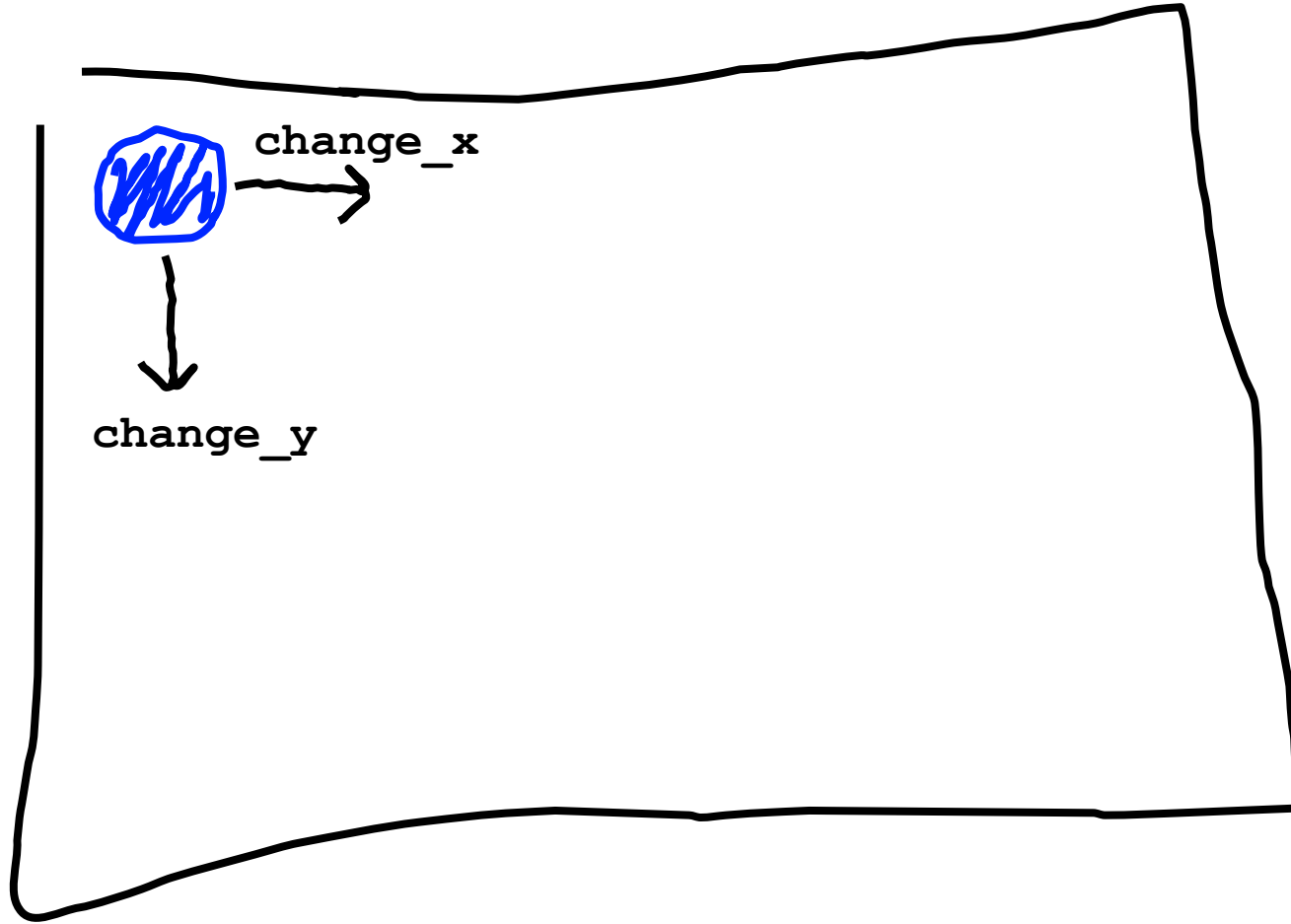


Key variable: how much the ball position change each heartbeat?

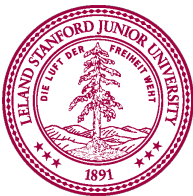


Bouncing Ball

First heartbeat

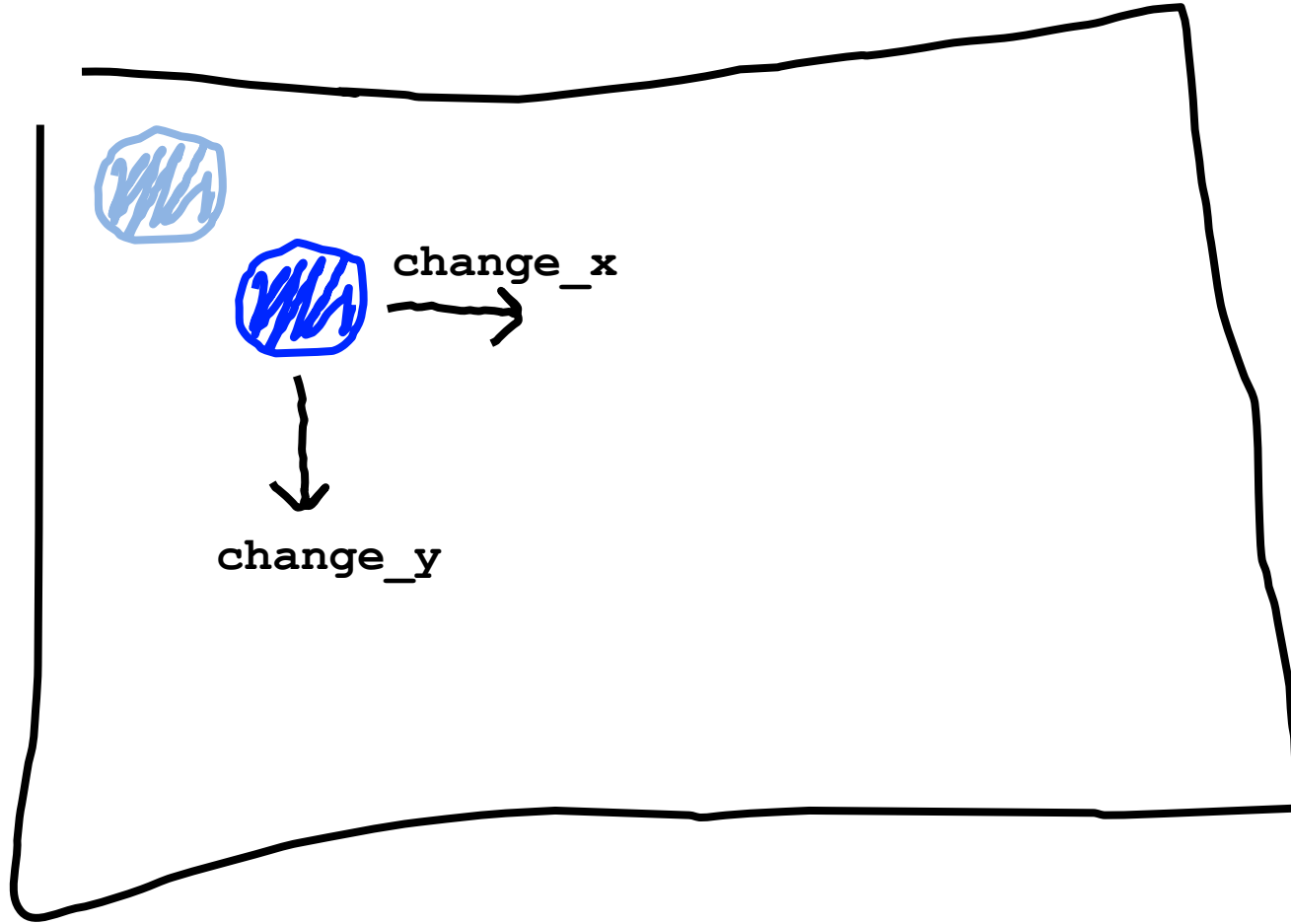


The **move** function takes in
a change in x and a change in y



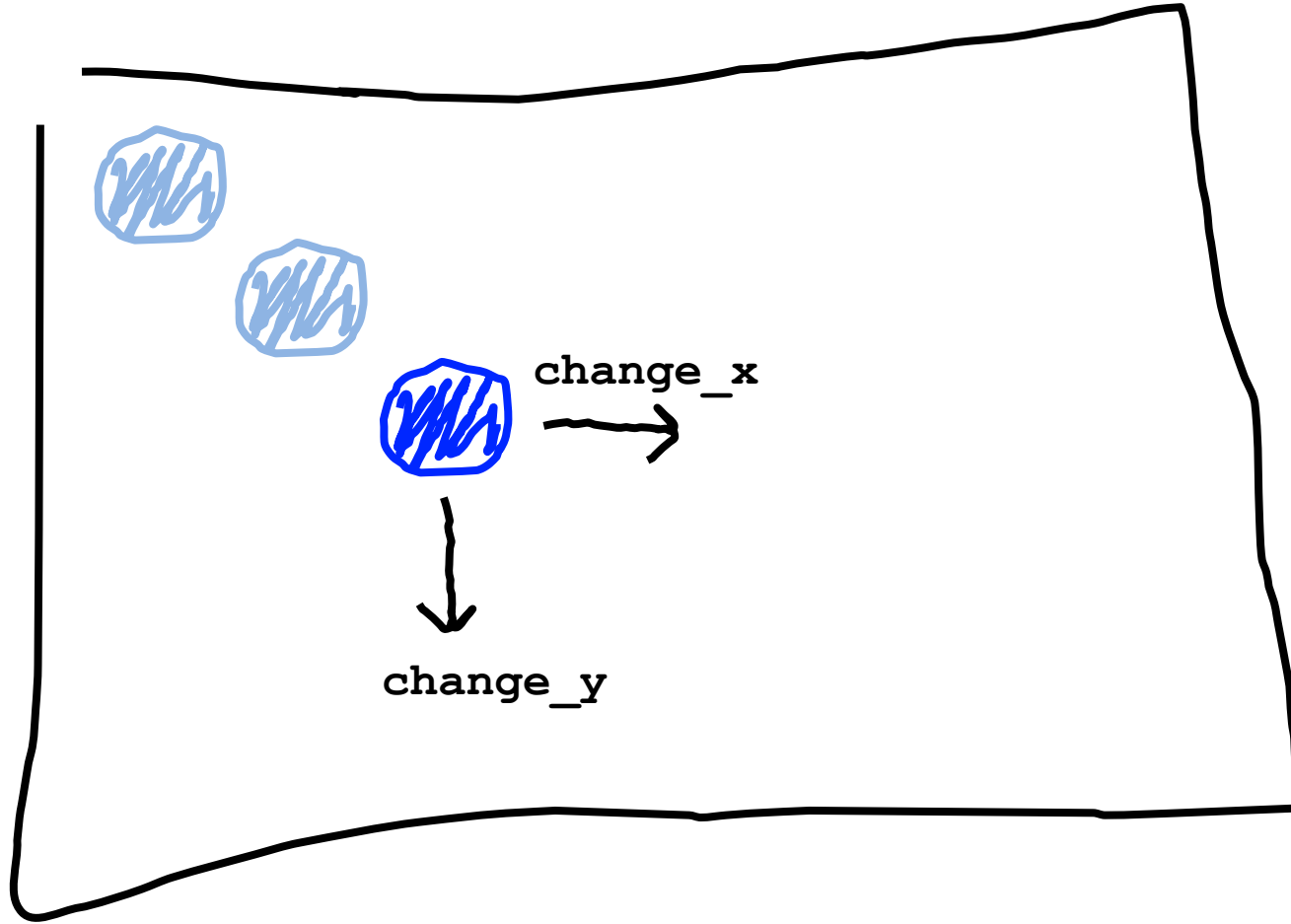
Bouncing Ball

Second heartbeat



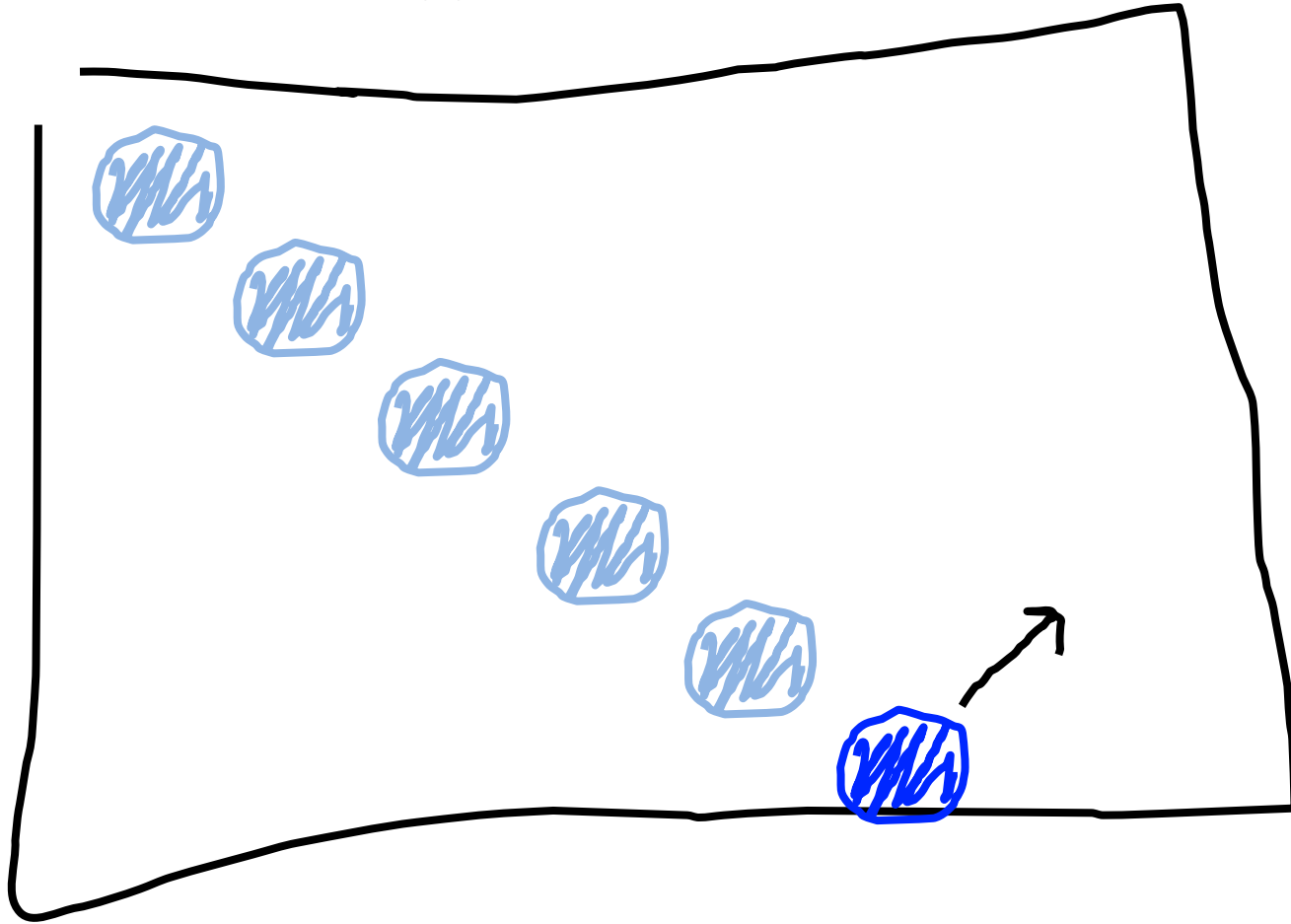
Bouncing Ball

Third heartbeat



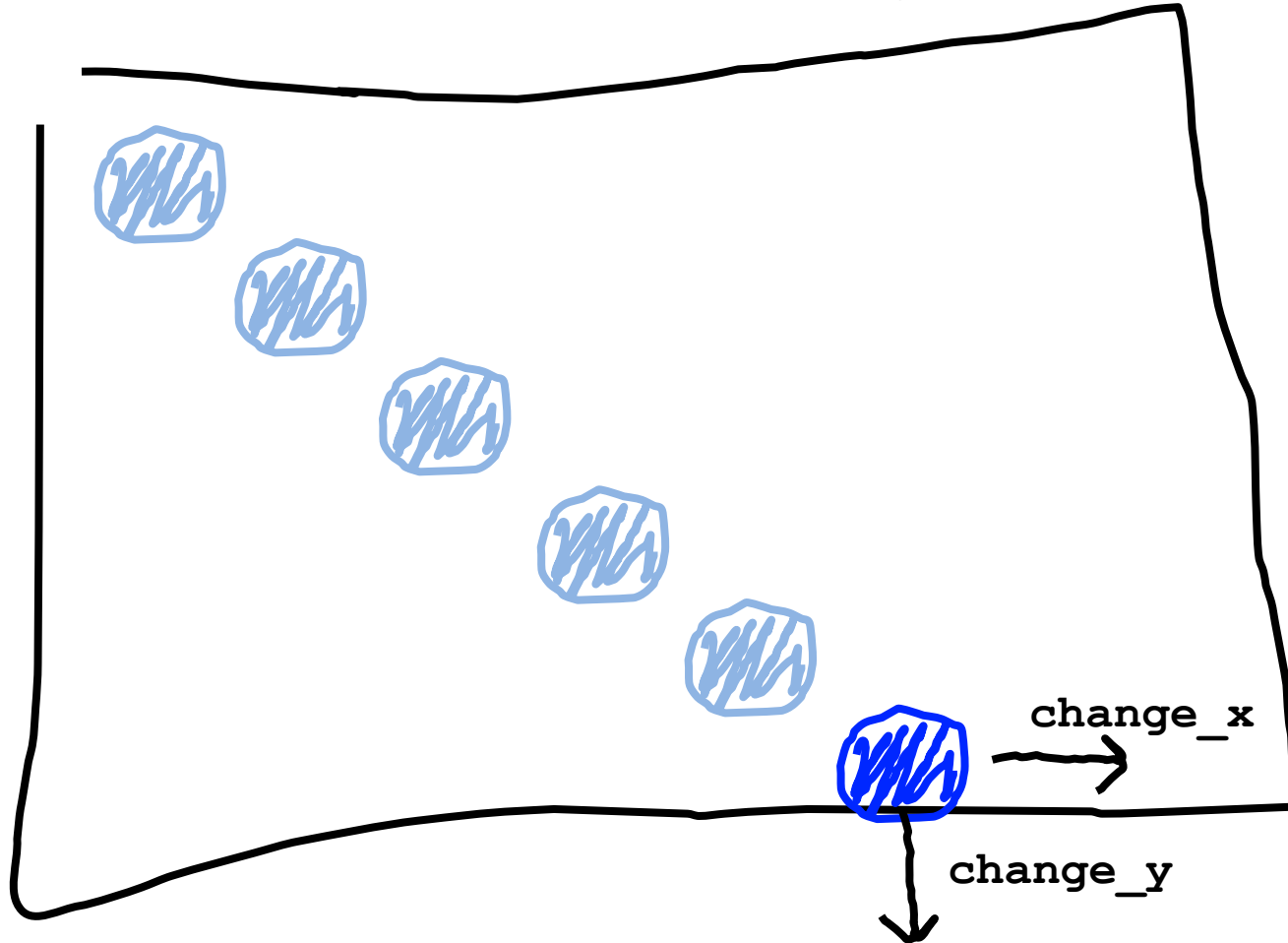
Bouncing Ball

What happens when we hit a wall?



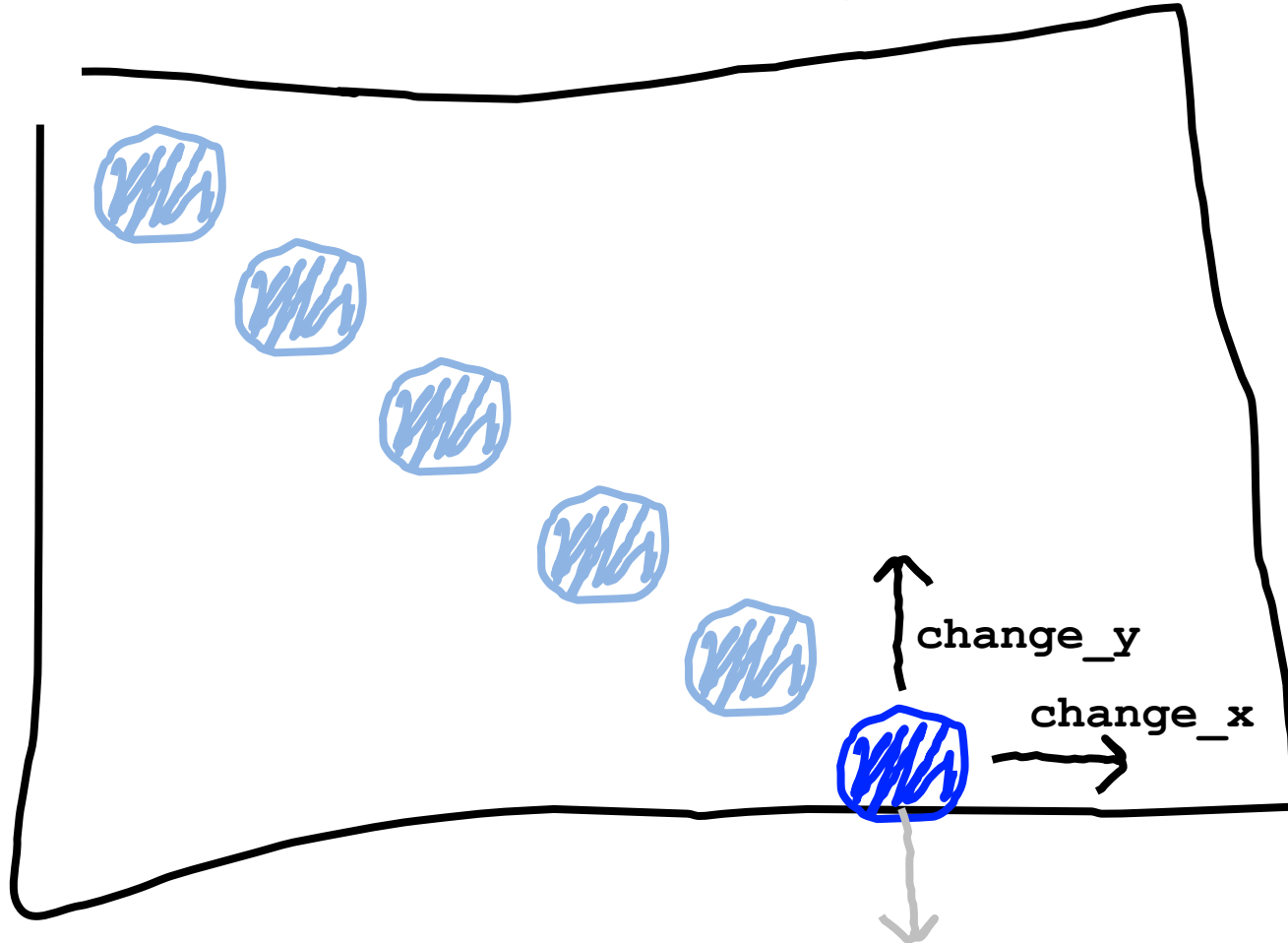
Bouncing Ball

We have this velocity

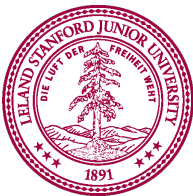


Bouncing Ball

Our new velocity

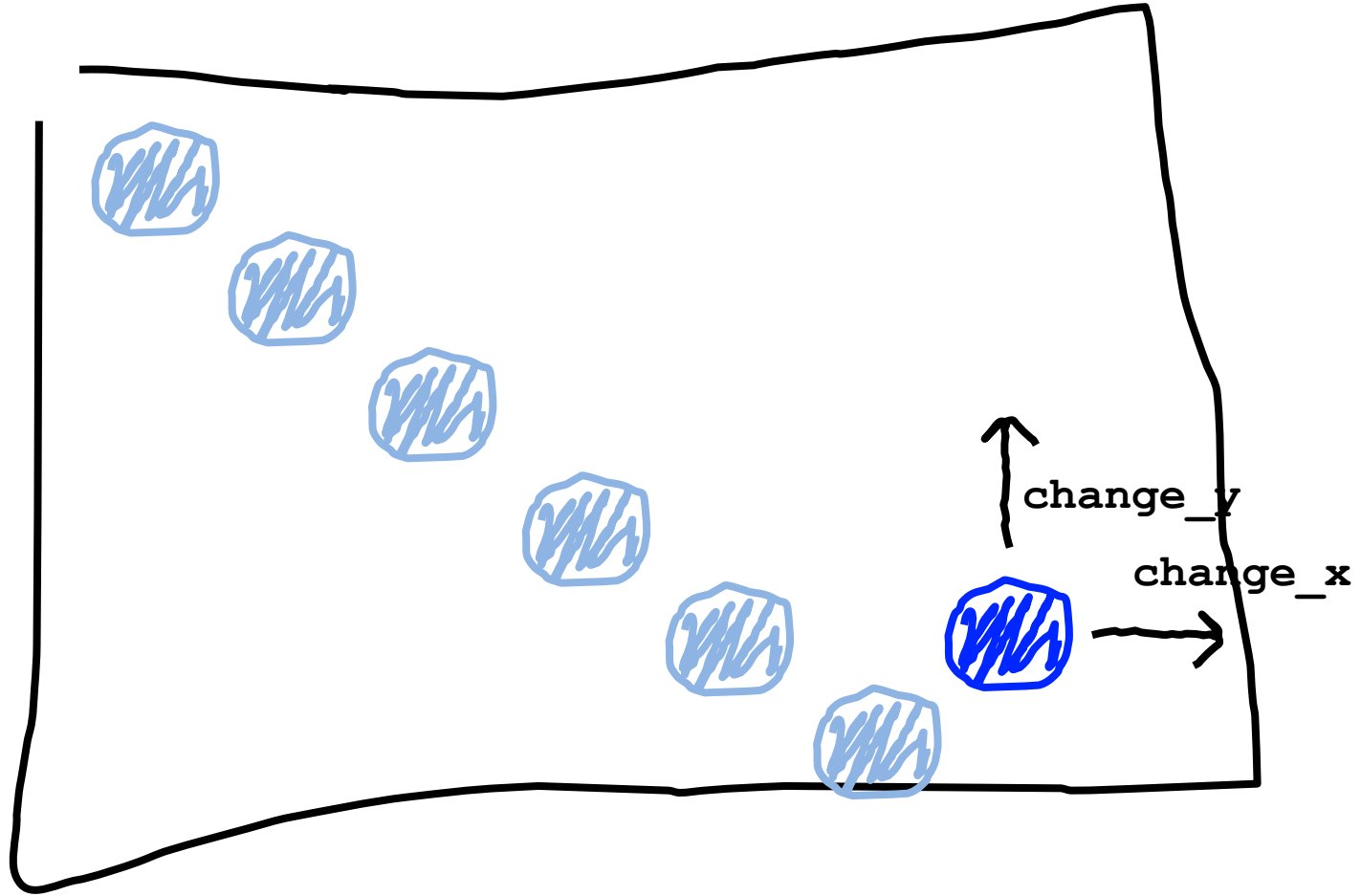


When reflecting vertically: $\text{change_y} = -\text{change_y}$



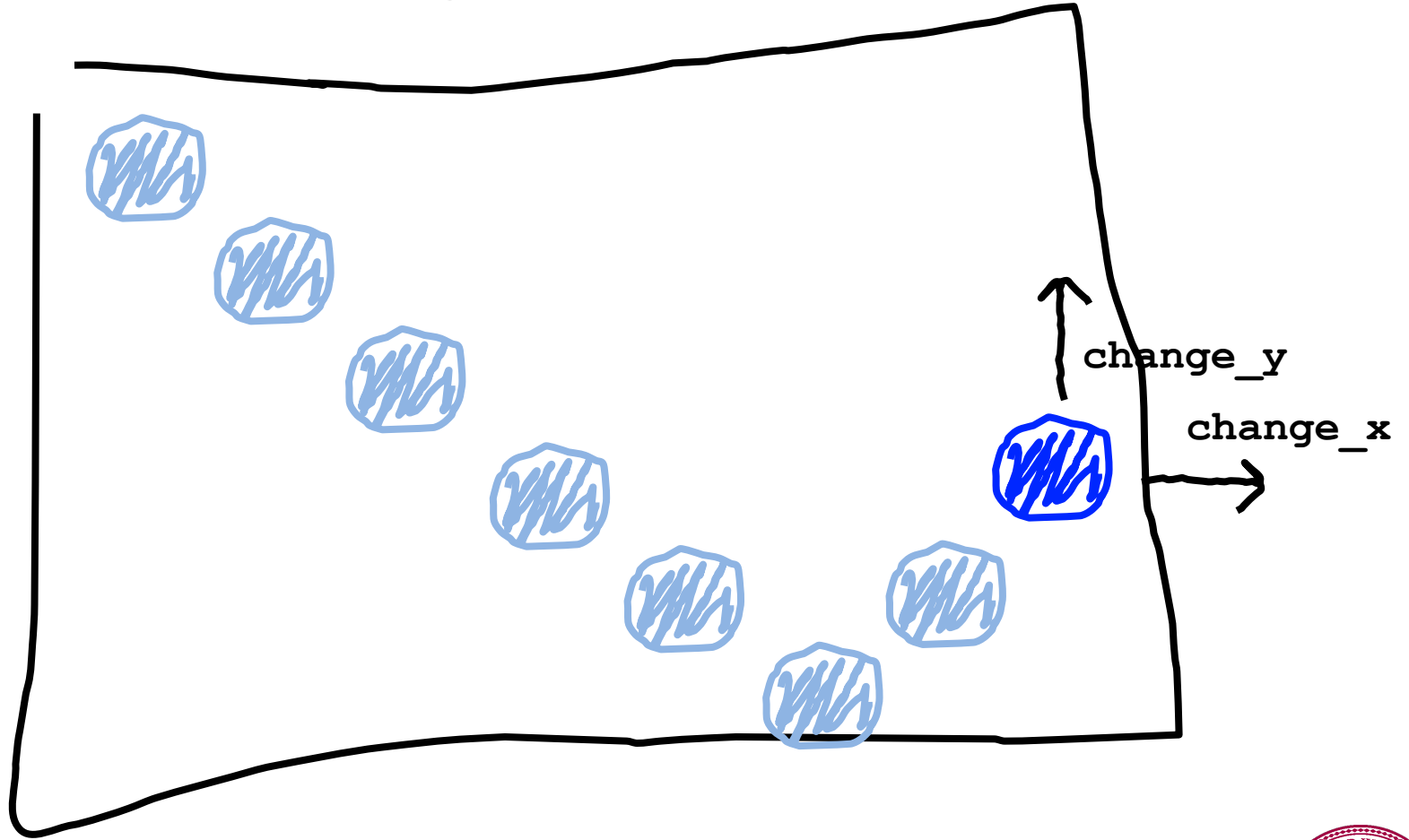
Bouncing Ball

Seventh heartbeat



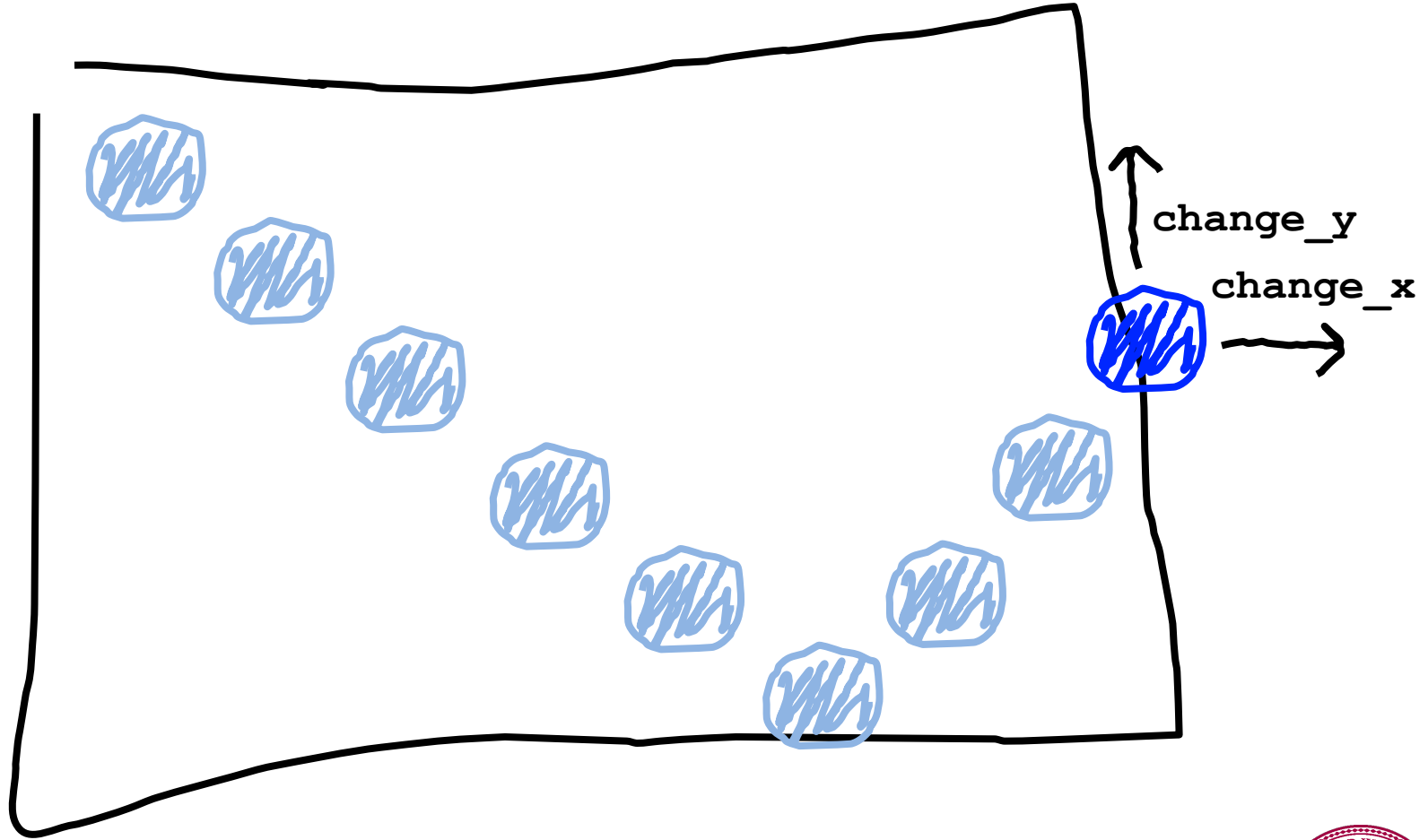
Bouncing Ball

Eighth heartbeat



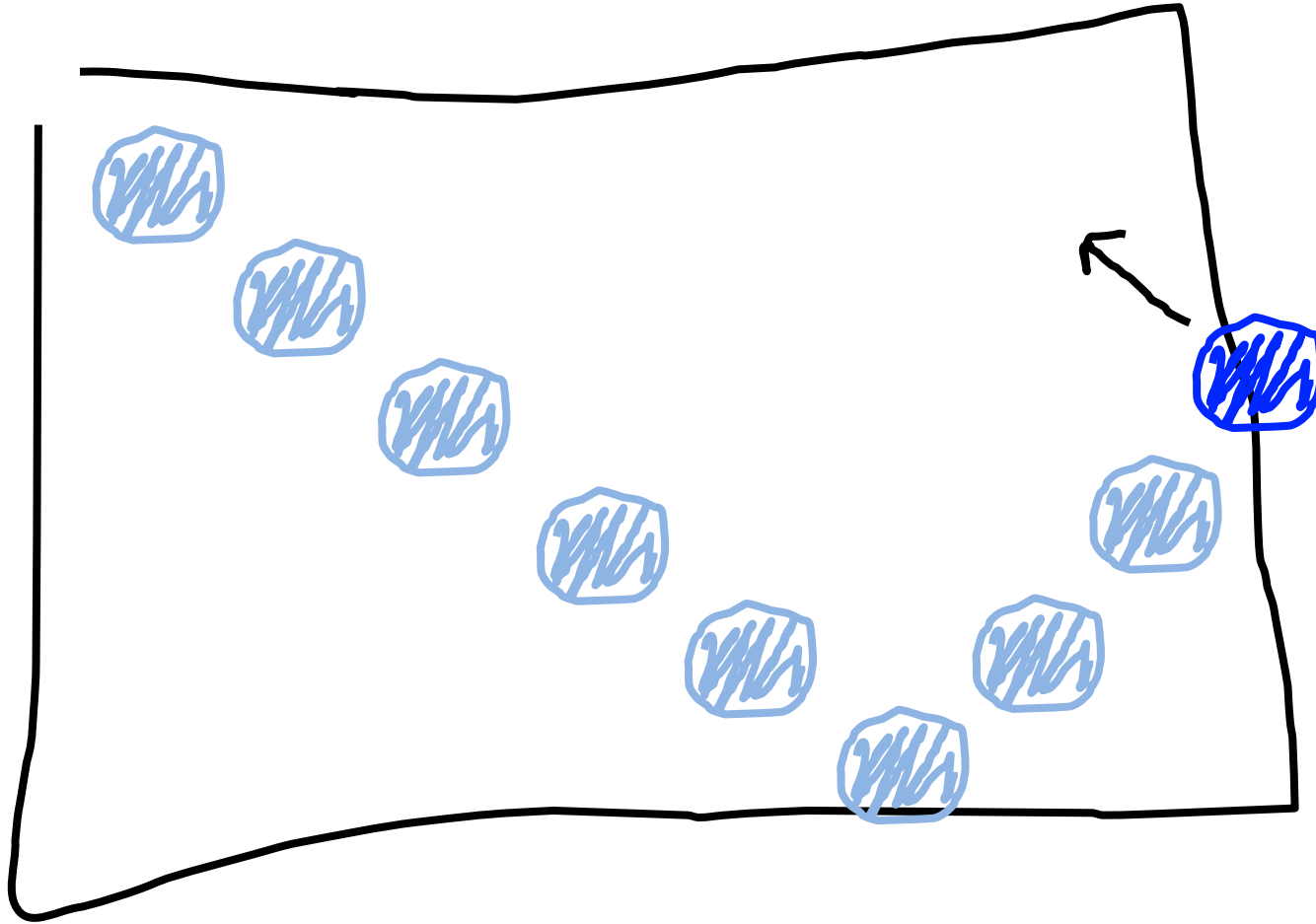
Bouncing Ball

Ninth heartbeat



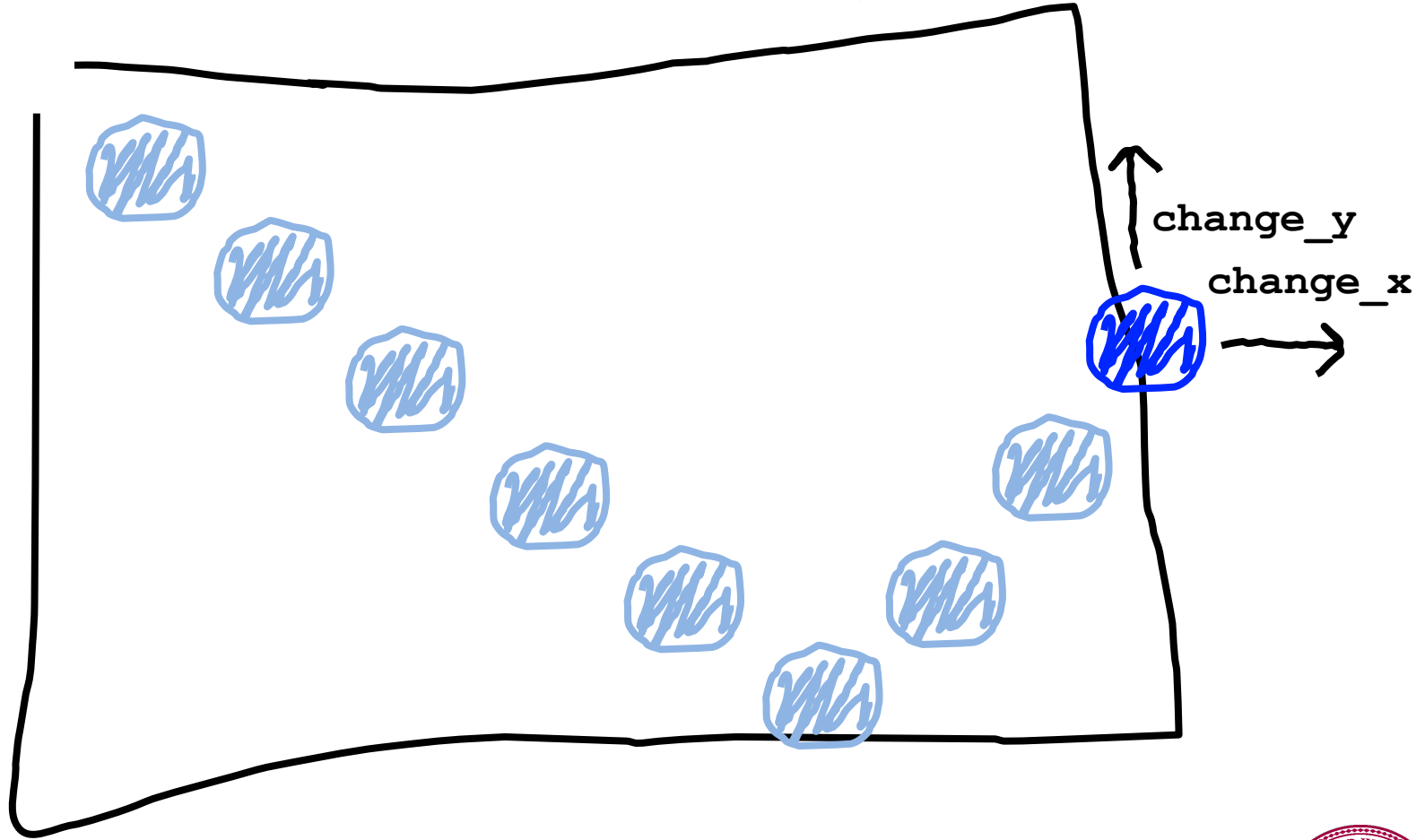
Bouncing Ball

We want this!



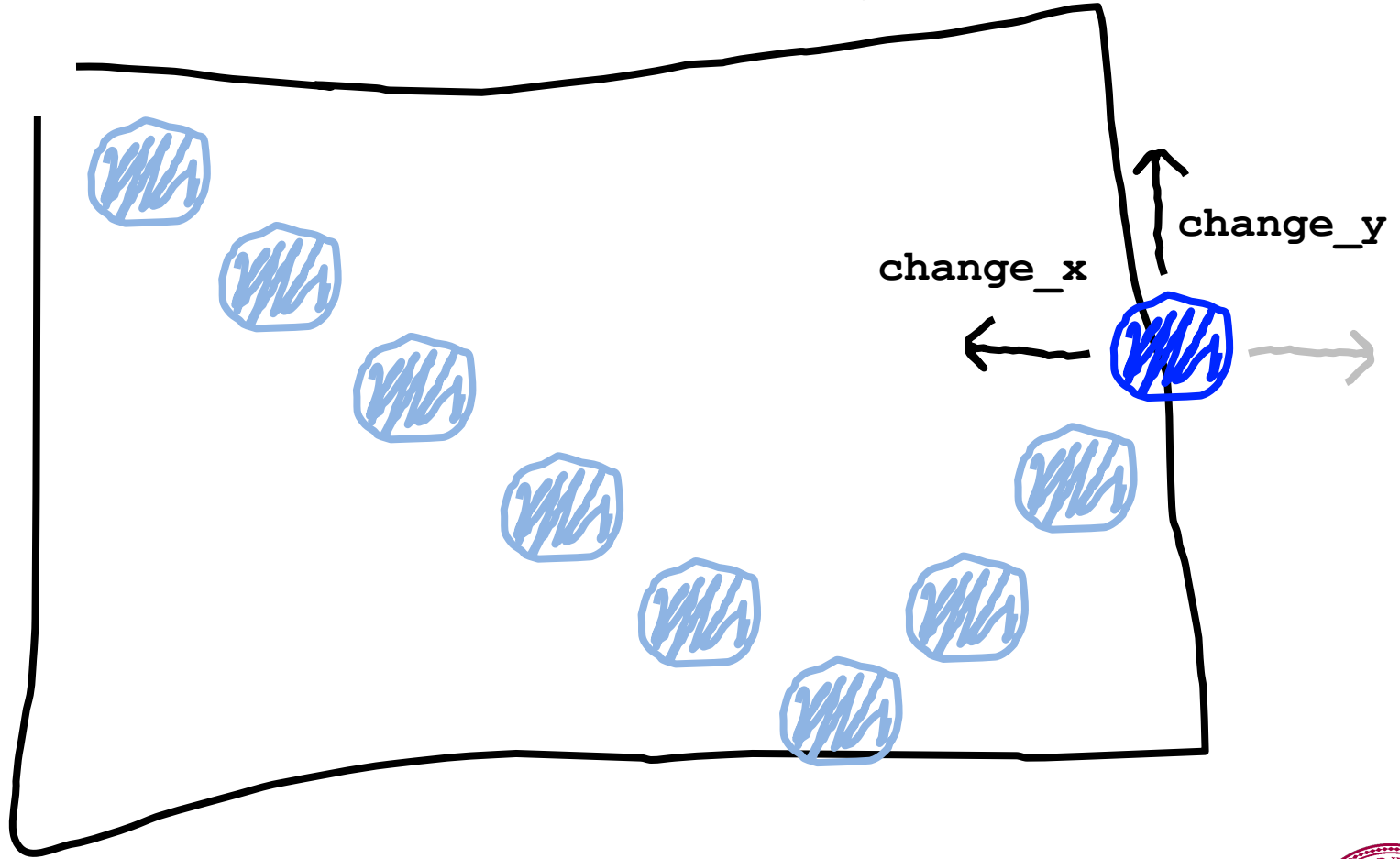
Bouncing Ball

This was our old velocity



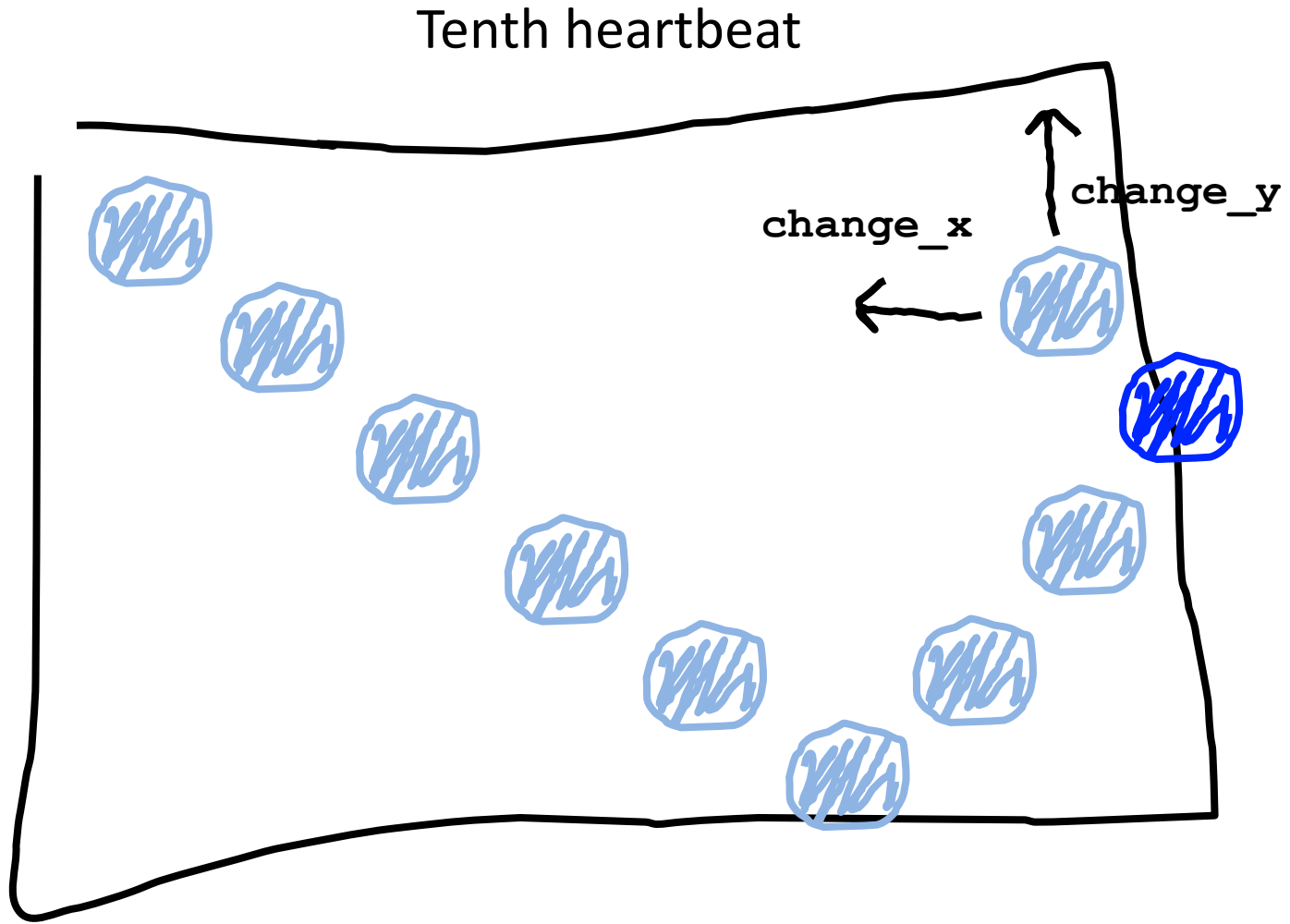
Bouncing Ball

This is our new velocity

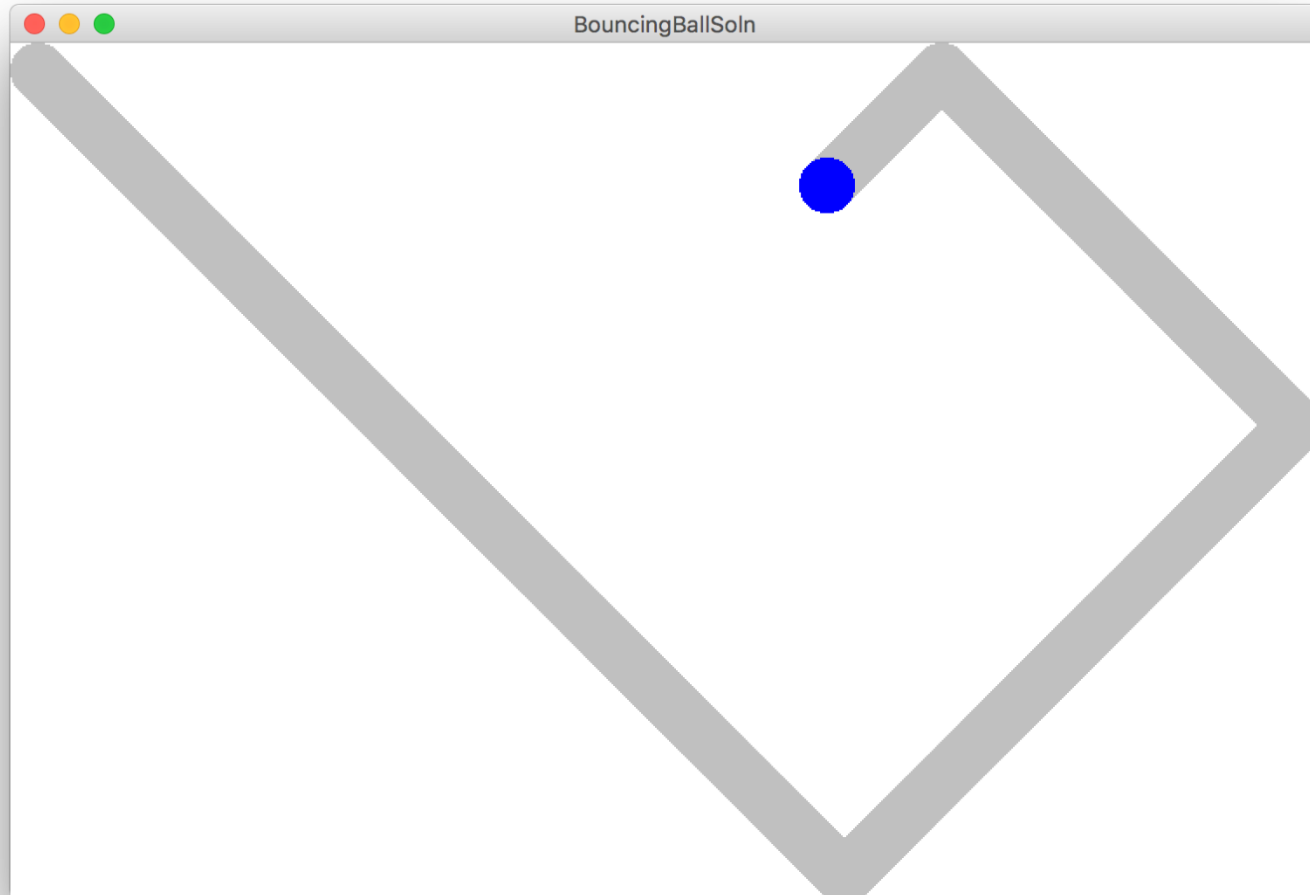


When reflecting horizontally: $\text{change_x} = -\text{change_x}$

Bouncing Ball



Bouncing Ball



Hold up!

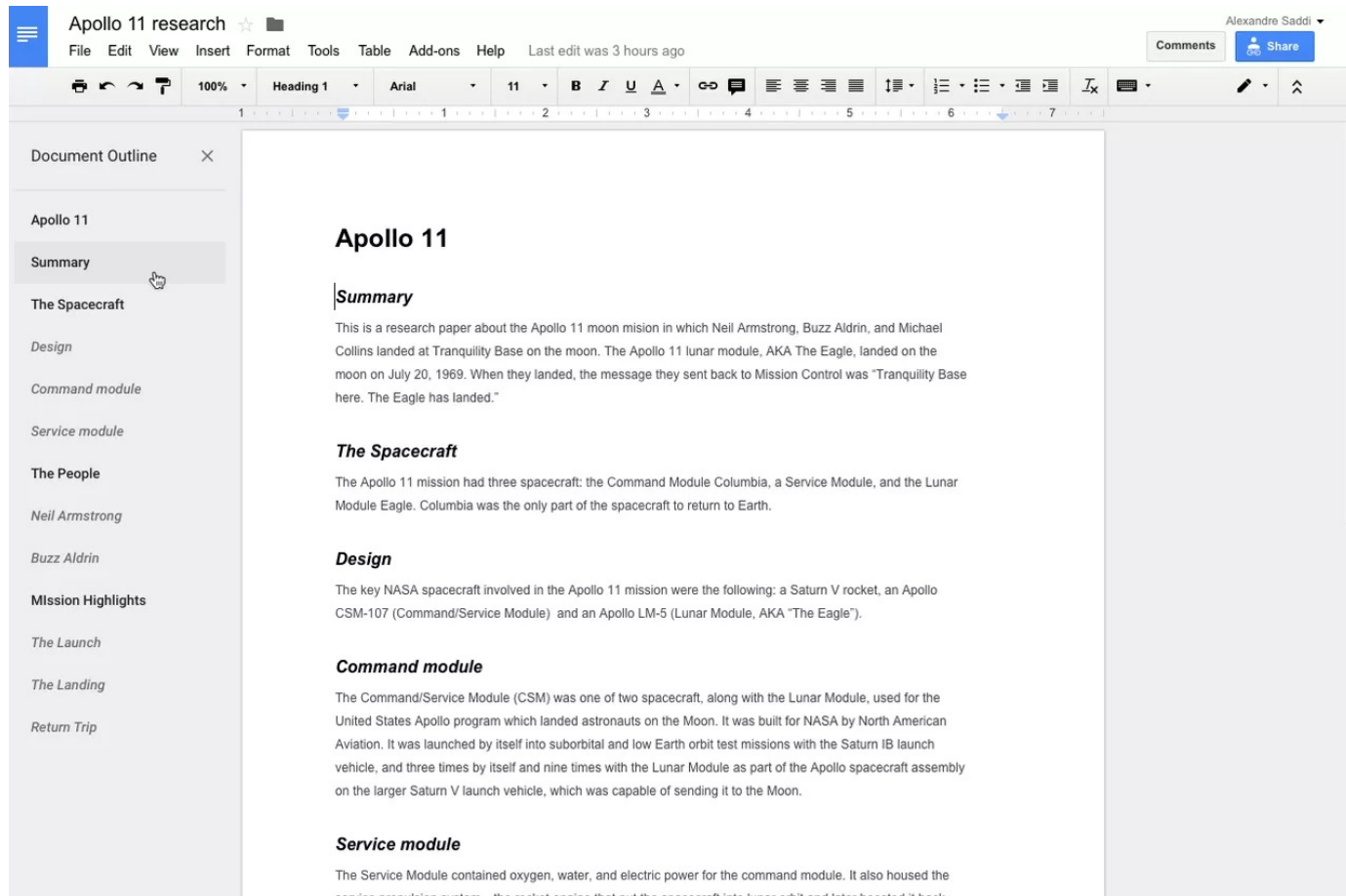
```
def make_ball(canvas) :
```

If you get a copy when you pass a parameter. Does this copy the canvas??!!

*Some Python variables (like Canvas) are stored using a reference. Which is like a **URL**. The URL gets copied when you pass the variable.*

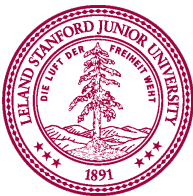


How do you share google docs?



The screenshot shows a Google Docs interface for a document titled "Apollo 11 research". The document is edited by "Alexandre Saddi" and was last edited 3 hours ago. The left sidebar contains a "Document Outline" with a tree structure: "Apollo 11" (parent), "Summary" (selected), "The Spacecraft", "Design", "Command module", "Service module", "The People", "Neil Armstrong", "Buzz Aldrin", "Mission Highlights", "The Launch", "The Landing", and "Return Trip". The main content area displays the "Summary" section, which includes a sub-header "Summary" and a paragraph: "This is a research paper about the Apollo 11 moon mission in which Neil Armstrong, Buzz Aldrin, and Michael Collins landed at Tranquility Base on the moon. The Apollo 11 lunar module, AKA The Eagle, landed on the moon on July 20, 1969. When they landed, the message they sent back to Mission Control was 'Tranquility Base here. The Eagle has landed.'". Below this are sections for "The Spacecraft", "Design", "Command module", and "Service module", each with a brief description of their role in the mission.

https://docs.google.com/document/d/1eBtnEill3KHe_fFS-kSAOpXqeSXpbfTTMImOgj6I9dvk/



```
def main():
```

```
    canvas = Canvas(...)
```

```
    make_ball(canvas)
```

```
def make_ball(canvas):
```

```
    canvas.create_oval( ... , 'blue')
```

stack

heap

main



```
def main():
```

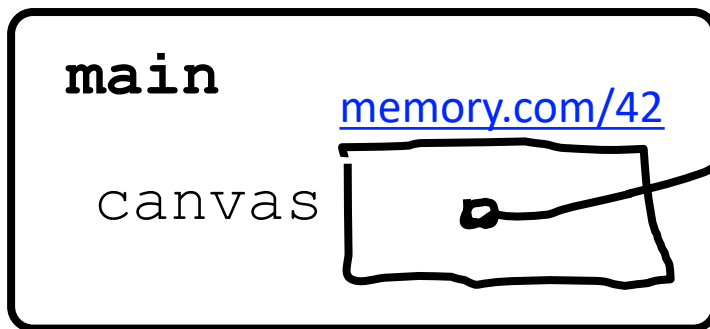
```
    canvas = Canvas(...)
```

```
    make_ball(canvas)
```

```
def make_ball(canvas):
```

```
    canvas.create_oval( ... , 'blue')
```

stack

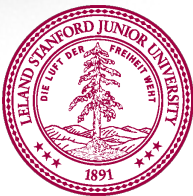
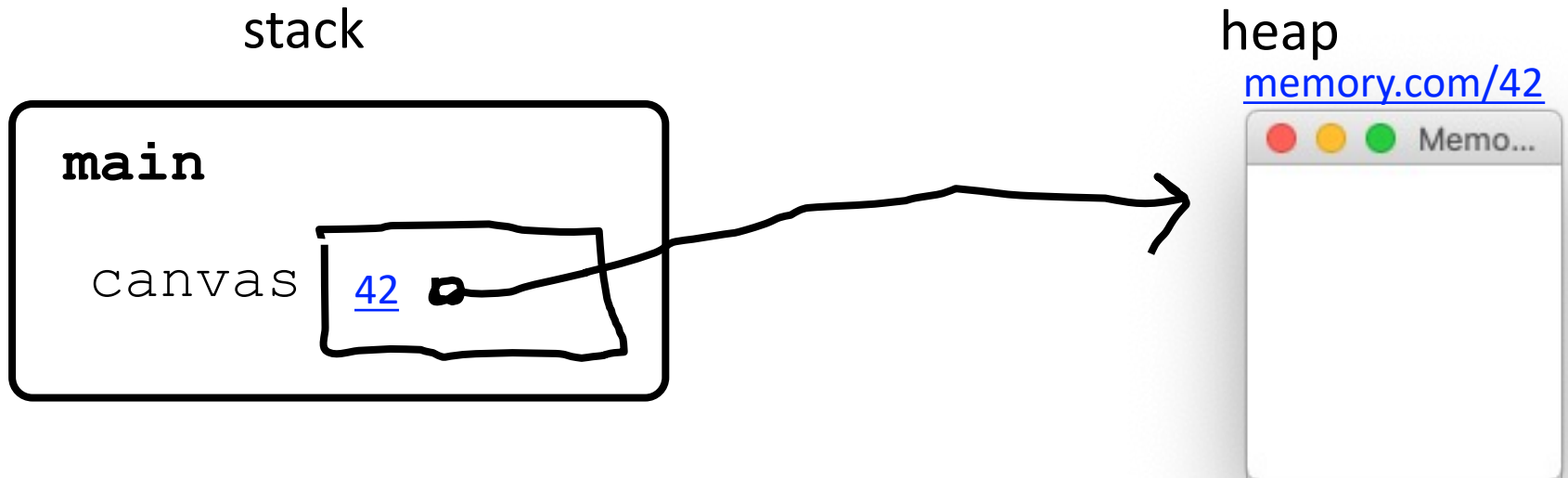


heap

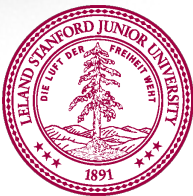
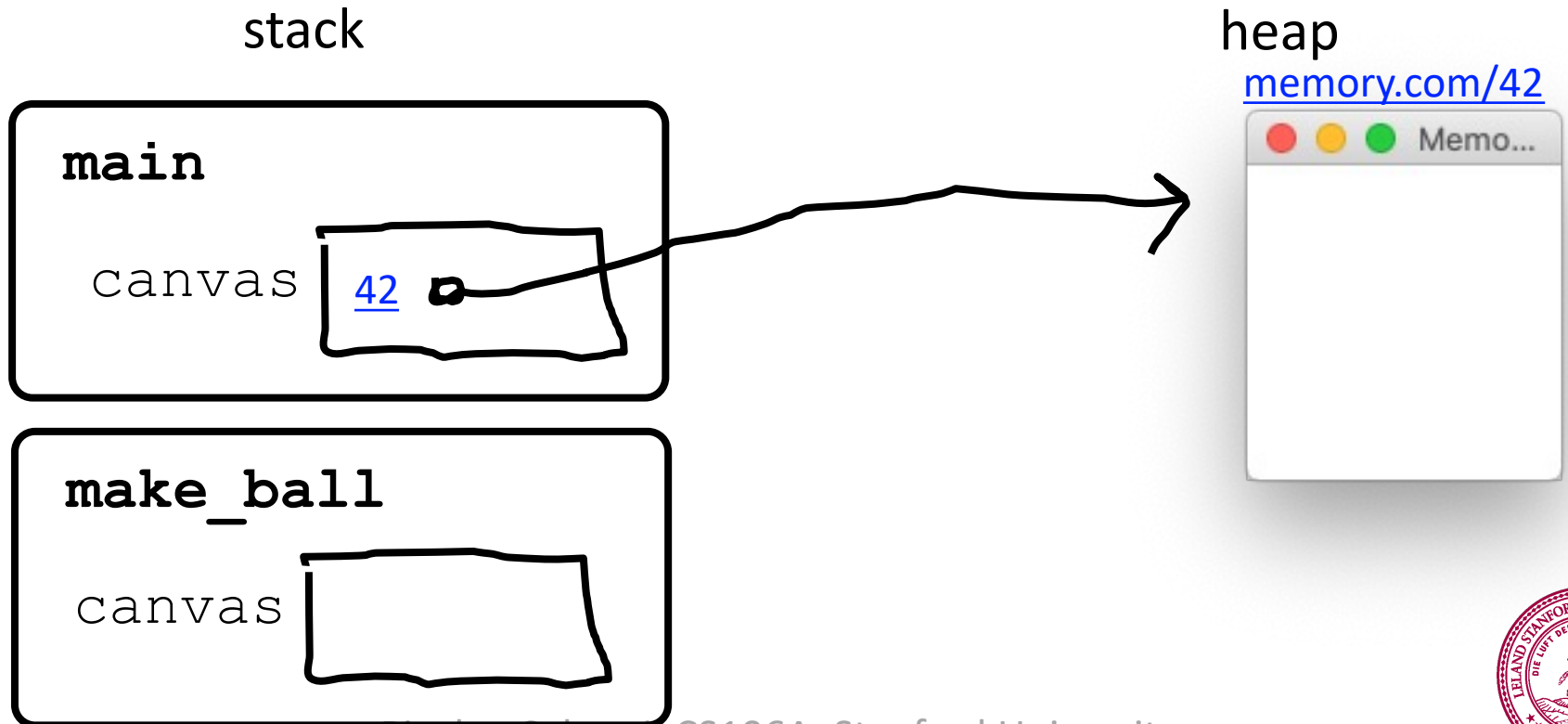
memory.com/42



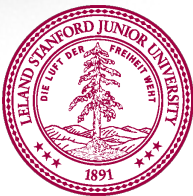
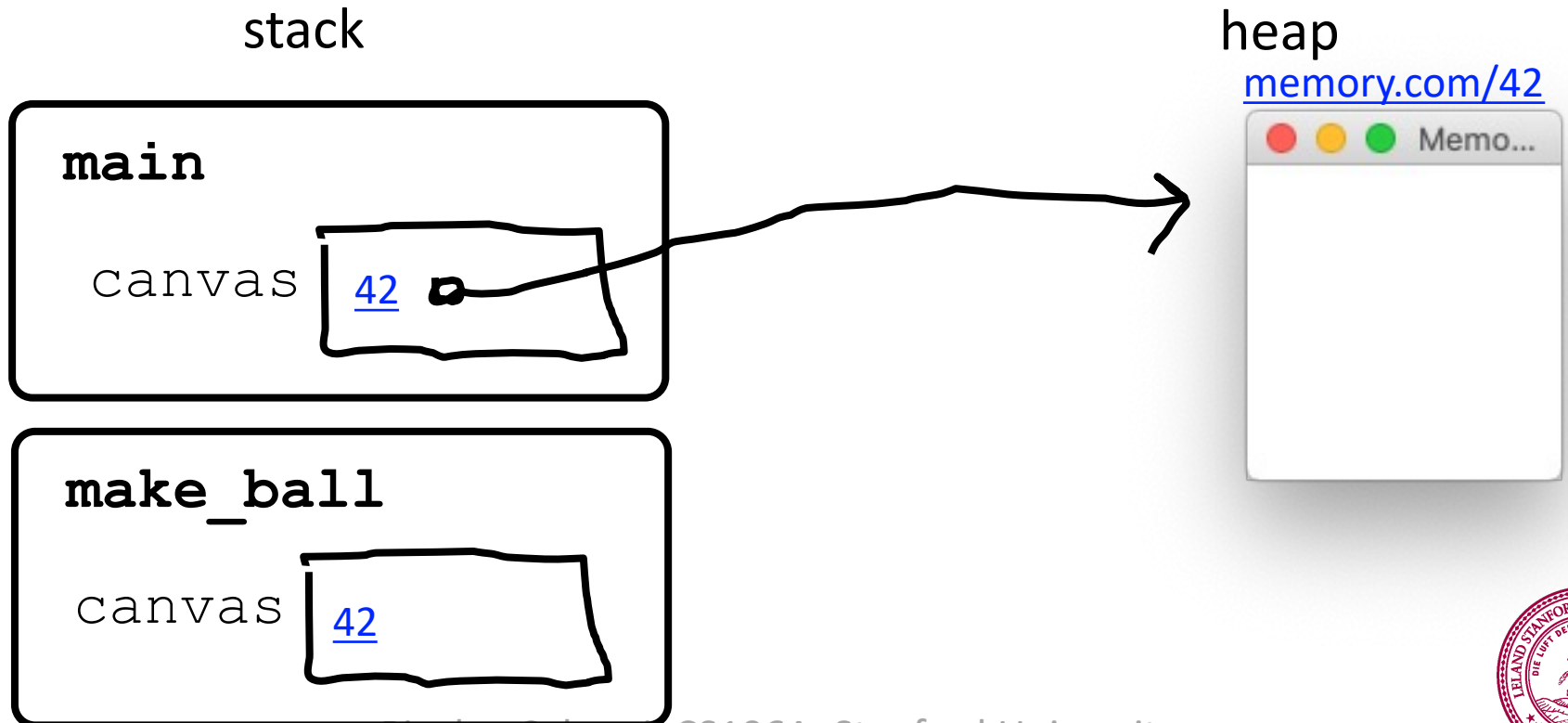
```
def main():  
    canvas = Canvas(...)  
    make_ball(canvas)  
  
def make_ball(canvas):  
    canvas.create_oval( ... , 'blue')
```



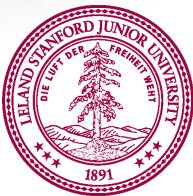
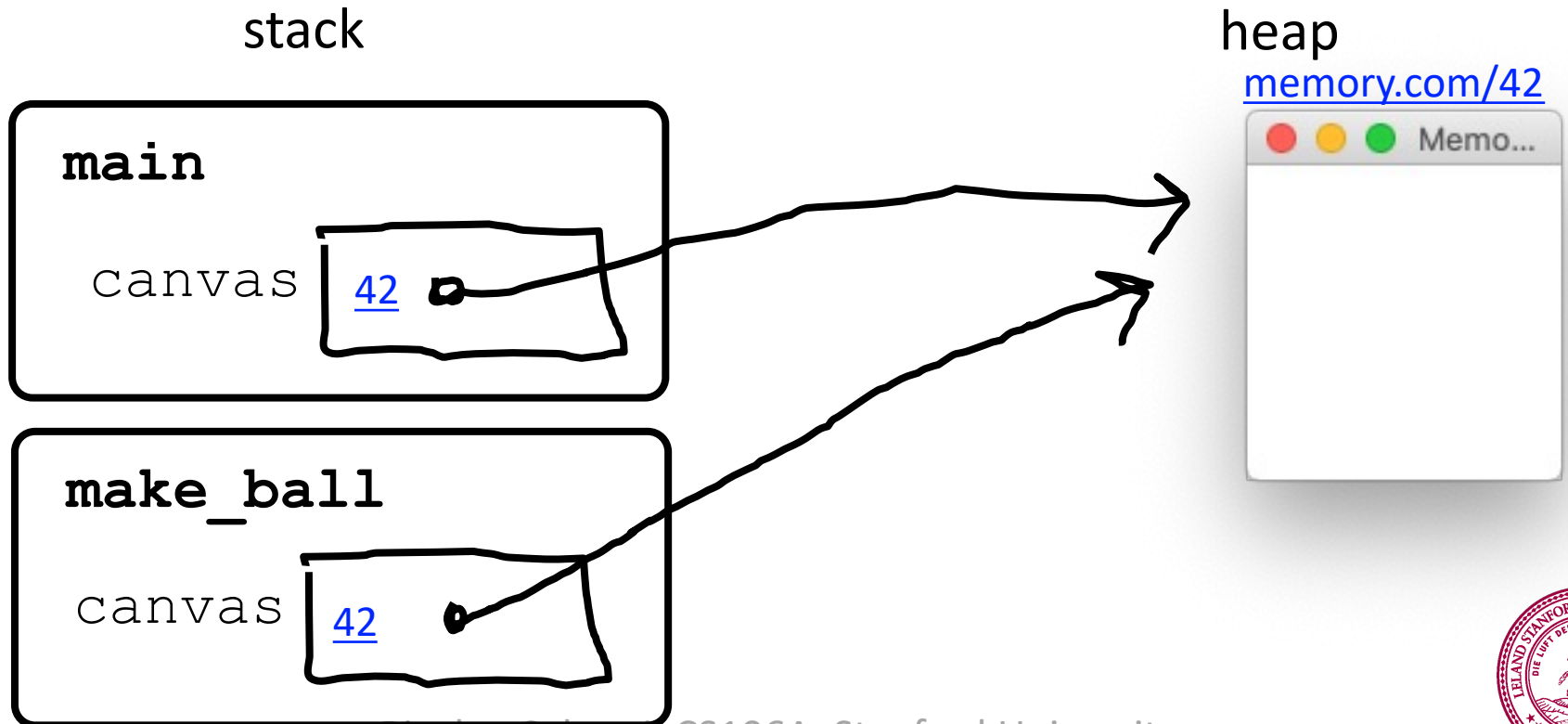
```
def main():  
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def make_ball(canvas):  
    canvas.create_oval( ... , 'blue')
```




```
def main():  
    canvas = Canvas(...)  
    make_ball(canvas)  
  
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    canvas.create_oval( ... , 'blue')
```

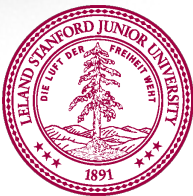
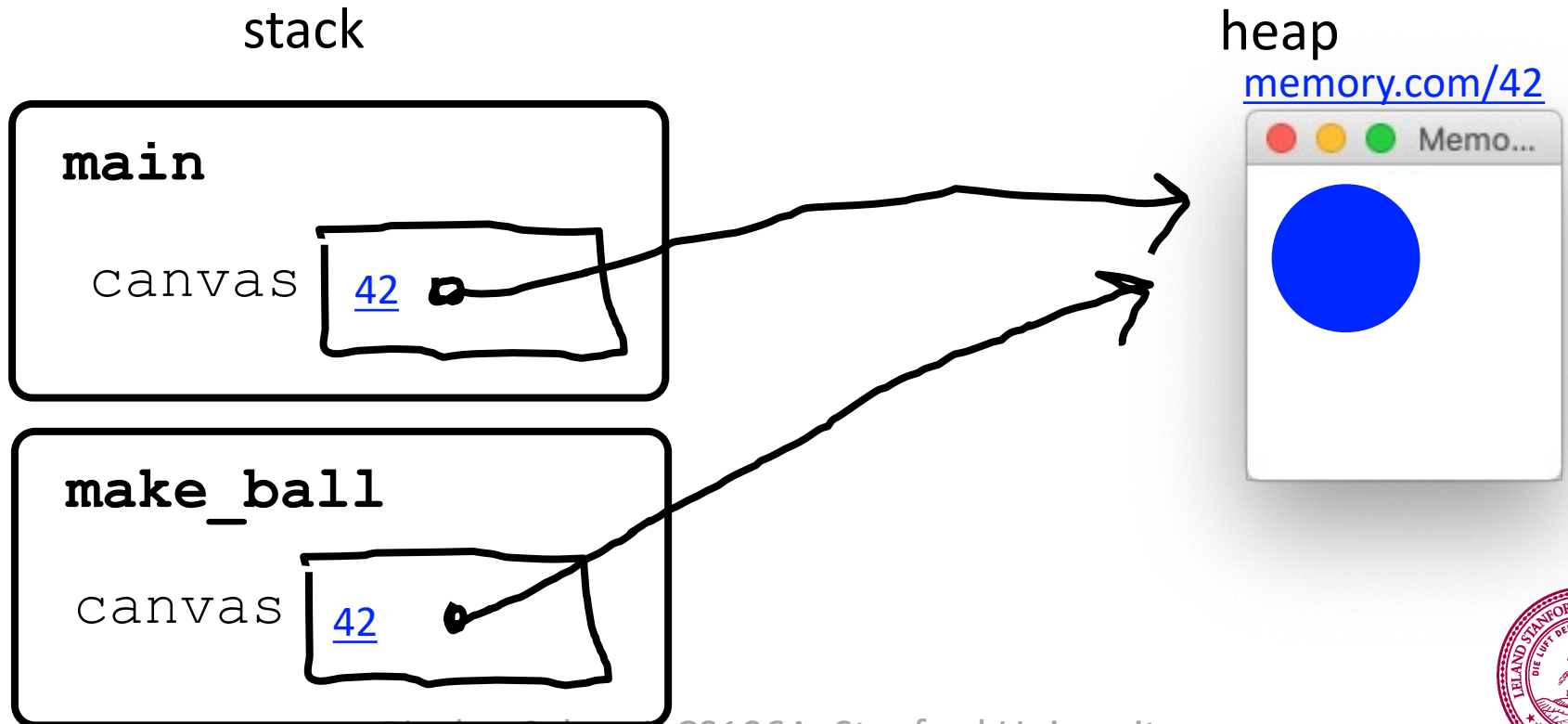


```
def main():  
    canvas = Canvas(...)  
    make_ball(canvas)  
  
def make_ball(canvas):  
    canvas.create_oval( ... , 'blue')
```



```
def main():  
    canvas = Canvas(...)  
    make_ball(canvas)
```

```
def make_ball(canvas):  
    canvas.create_oval( ... , 'blue')
```

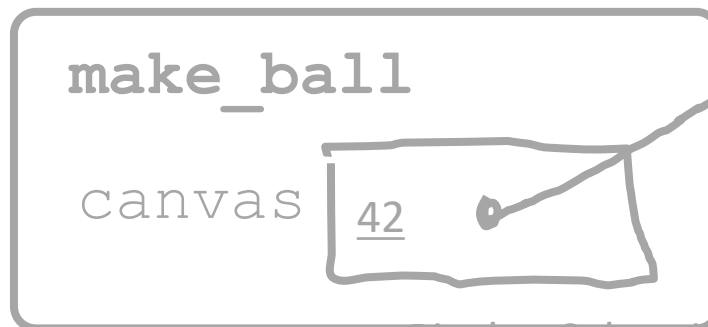
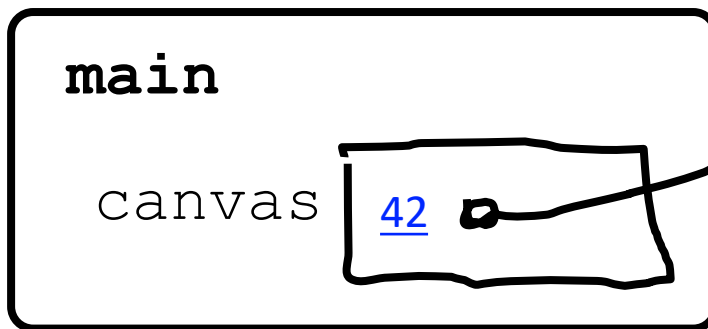


```
def main():  
    canvas = Canvas(...)  
    make_ball(canvas)
```

```
def make_ball(canvas):  
    canvas.create_oval( ... , 'blue')
```

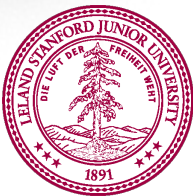
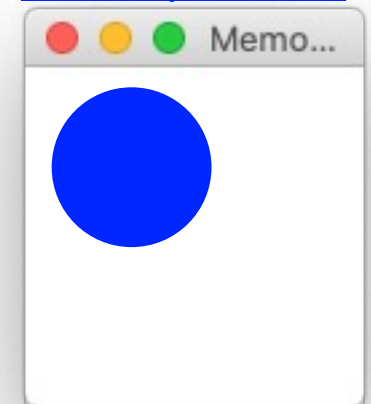


stack

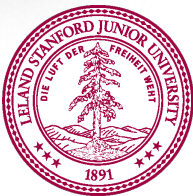
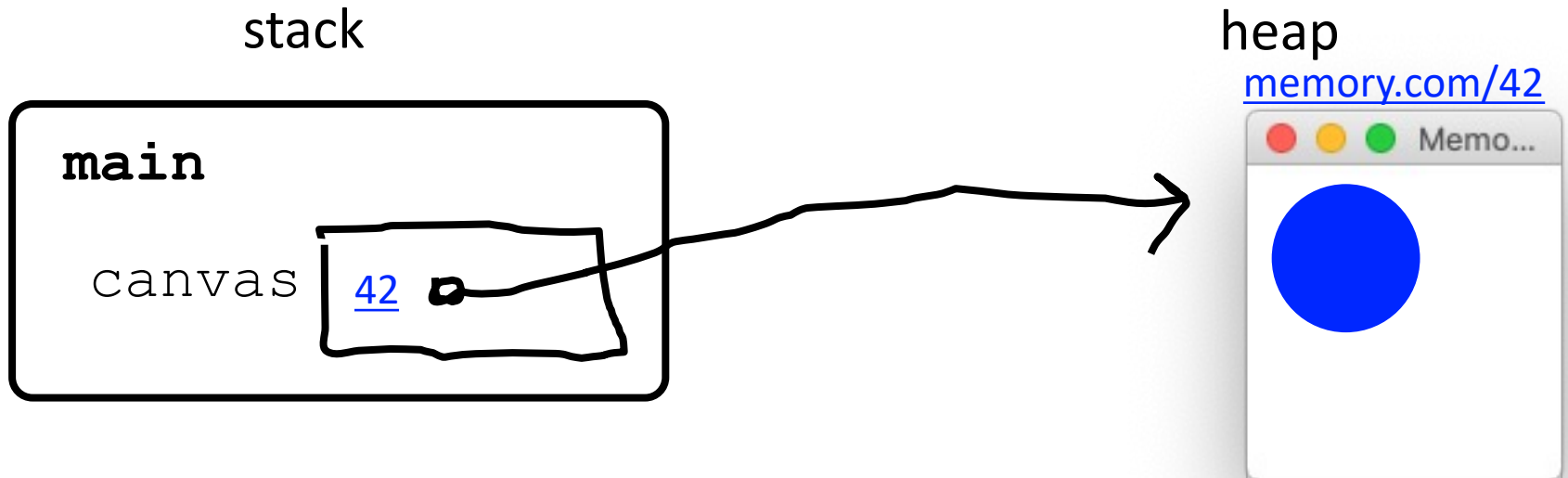


heap

memory.com/42



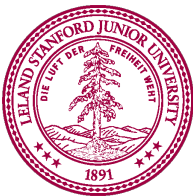
```
def main():  
    canvas = Canvas(...)  
    make_ball(canvas)  
  
def make_ball(canvas):  
    canvas.create_oval( ... , 'blue')
```





When passing variables,
some act just like you are
passing a URL.

That allows functions to
modify the variable



Special Graphics Functions

get the x and y location of the mouse

```
mouse_x = canvas.get_mouse_x()  
mouse_y = canvas.get_mouse_y()
```

move shape to some new coordinates

```
canvas.moveto(shape, new_x, new_y)
```

move shape by a given change_x and change_y

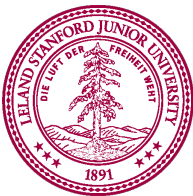
```
canvas.move(shape, change_x, change_y)
```

get the coordinates of a shape

```
top_y = canvas.get_top_y(shape)  
left_x = canvas.get_left_x(shape)
```

return a list of elements in a rectangle area

```
results = canvas.find_overlapping(left_x, top_y, right_x, bottom_y)
```



Let's track the mouse!
(Fun for things like games)